



University of Venda
Creating Future Leaders

Faculty of Science, Engineering and Agriculture

**DEPARTMENT OF MATHEMATICAL AND
COMPUTATIONAL SCIENCES**

MAT1141

Differential Calculus

**Compiled by Dr VT Makhoshi
January 2026**

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Learning Guide

1 The Lecturer

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2 Foreword

We provide you with important information regarding this module, Differential Calculus. It is compulsory that you attend all contact and virtual lectures, pay attention and ask questions where you do not understand. Always keep in mind that the module lecturer is your primary source of information. Where possible, form study groups and invest at least one hour per day in studying and practicing calculus.

3 General Information on the Module

- NQF Level: 5
- NQF Credits: 8
- Type of module: 1st Semester
- Notional Hours: 80

4 Purpose of the Module

To provide you with a firm foundation which will support some of your future learning. This will be done by studying the limit of a function and a special class of functions called continuous functions. We will use the notion of the limit of a function to introduce the derivative of a function and invest most of our time exploring various rules of differentiation, culminating our study by discussing the applications of differential calculus in solving real life problems.

4.1 Course Outline

Introductory concepts: Real numbers, review of elementary mathematics, definition of a function, functions and their graphs, trigonometric functions, exponential functions, logarithmic functions, hyperbolic and inverse trigonometric functions. Limits and continuity,

The Derivative of elementary functions: Rules of differentiation; basic rules of differentiation, product rule, quotient rule, derivatives of trigonometric, exponential, logarithmic, hyperbolic and inverse trigonometric functions. The Chain Rule, Implicit differentiation. Differentials and linear approximations,

Applications of the derivative: L'Hôpital's rule, Taylor and Maclaurin polynomials, Extrema of functions, The mean-value theorems for derivatives, Increasing and decreasing functions, concavity and inflection points, limits involving infinity and asymptotes, curve sketching.

5 Outcomes

Critical Cross Field Outcomes

Students should be able to:

- i. Identify and solve problems
- ii. Work in a team
- iii. Reflect on and explore effective learning strategies

Specific Outcomes At the end of this module, students should be able to:

- a. Define and interpret elementary functions.
- b. Give an intuitive explanation of the process of taking a limit and compute basic limits of functions at a point numerically and algebraically using appropriate techniques.
- c. Verify the value of a limit of a function at a point using the precise definition of the limit,
- d. Define continuity of a function at a point and in an interval. Find and classify points of discontinuity for functions. Understand the consequences of the Intermediate Value Theorem for continuous functions.
- e. Explain the relation between limits and the process of differentiation, state the definition of the derivative as the limit of the difference quotient and compute the derivative of a simple function directly using limits.
- f. Derive the expression for the derivative of elementary functions from the limit, differentiate composite functions using the chain rule, perform implicit differentiation on relations, compare and contrast the ideas of continuity and differentiability.

- g. Use the differential to determine the error of approximations. Understand the consequences of Rolle's Theorem and the Mean Value Theorem for differentiable functions. Use differentiation for curve sketching and analysis thereof, identify the extrema of a function on an interval and classify them as minima, maxima or saddles using the first derivative test. Compute Taylor's polynomial of specified degree about a given point for a given function and use basic Taylor series for computation. Use L'Hôpital's rule to evaluate limits.

6 Assessments

Formative assessment

Students may be divided into groups. Should that be the case, each student will be expected to belong to exactly one group.

2 (major) tests may be written per semester and the marks obtained will be used to calculate the student's Semester mark.

Calculation of Semester Mark: 40 % (Class tests and Assignments) and 60% Major Tests.

To qualify to write the examination a student must get an average of at least 40% in the formative assessments.

Summative Assessment A three hour examination paper out of 100 marks is written at the end of the semester.

Pass Requirements The final Pass Mark is 50%

Final Mark Compilation: Semester Mark + Examination Mark 60% + 40% A student must get a **subminimum mark of 40% in the examination.**

Assessment Dates

TEST 1 28 Feb 2026 11h00-13h00

TEST 2 18 Apr 2026 11h00-13h00

7 Support Resources

Recommended Books

- Stewart, James (2021), Calculus. 9th Edition. Brooks/ Cole Publishing Company. (Any edition will do)
- Soo T, Tan. (2010). Calculus. Brooks/ Cole Publishing Company. (Any edition will do)
- Salas; Hille; Etgen(2007) Calculus, One and Serveral Variables. John Wiley and Sons,Inc (Any edition will do)
- Thomas B, George; Wier D, Maurice and Hass R, Joel (2010). Thomas' Calculus. Pearson Education. (Any edition will do)
- Any other standard Calculus text book.

8 Rules

8.1 Rules and procedure regarding deferred (aegrotat) assessment opportunities

0.8.1 A student may be granted a special assessment opportunity if he/she applies in writing within 5 working days subsequent to the original date of assessment missed, this must be handed over to your own lecturer in person where possible or otherwise via email.

0.8.2 Reason for absence

- Illness: Medical certificate, completed by the medical practitioner with an official stamp must accompany the application as proof.
- Compassionate reasons: (Immediate family only) Relevant documentation such as death certificate etc. and a letter detailing circumstances and the student's details must accompany the application.
- Legitimate reasons: An affidavit signed and stamped by a commissioner of oaths, accompanied by a letter, detailing the specific circumstances and the student's details.

The discretion of the lecturer will play a major role in the making of the final decision.

8.2 Conduct

- ALL CONTACT and VIRTUAL LECTURES are COMPULSORY.
- Noise (or any form of disturbance) will NOT be tolerated during contact or virtual lectures.
- Students are to be punctual at all times.
- Due dates for tasks must be respected.
- Test dates and venues will be announced during lectures, on Moodle and/or shared via Whatsapp.
- A student who copies other student's work or allows others to copy (tests, assignments, etc.) his/her work will be given a zero.
- Students are **strongly discouraged** from hiring people to write assignments for them.
- Cell phones are strictly prohibited during tests and examinations.
- Students are expected to be honest and responsible at all times.

"When you have eliminated the impossible, whatever remains, however improbable, must be the truth." - **Sir Arthur Conan Doyle**

1 Preliminaries

1.1 Review of Elementary Mathematics

We give a review of the terminology, notation, and some formulas of elementary mathematics.

Sets

A set is a collection of distinct objects. The objects in a set are called the *elements* or *members* of the set. We shall denote sets by capital letters $A, B, C, \dots$ and use lowercase letters $a, b, c, \dots$ to denote elements. We say

the object x is in the set, $A \quad x \in A$
the object x is not in the set, $A \quad x \notin A$
the set of all x which satisfy the property, $P \quad \{x : P\}$
the empty set, $\emptyset$.

Real Numbers $\mathbb{R}$

natural numbers $\mathbb{N} \quad 1, 2, 3, \dots$
positive integers $\mathbb{Z}^+ \quad 1, 2, 3, \dots$
integers $\mathbb{Z} \quad \dots -3, -2, -1, 0, 1, 2, 3, \dots$
rational numbers $\mathbb{Q}$, such that $\frac{p}{q}$, $p, q \in \mathbb{Z}$, $q \neq 0$
irrational numbers $\mathbb{R} \setminus \mathbb{Q} \quad \sqrt{2}, \pi, e$

Order Properties

- i. Either $a < b$, $b < a$, or $a = b$.
- ii. If $a < b$ and $b < c$ then $a < c$.
- iii. If $a < b$, then $a + c < b + c$, $\forall c \in \mathbb{R}$.
- iv. If $a < b$ and $c > 0$, then $ac < bc$.
- v. If $a < b$ and $c < 0$, then $ac > bc$.

Boundedness

A set S of real numbers is said to be:

- i. *Bounded above* if there exists $M \in \mathbb{R}$ such that $x \leq M \quad \forall x \in S$, such a number M is called an *upper bound* for S .
- ii. *Bounded below* if there exists $m \in \mathbb{R}$ such that $m \leq x \quad \forall x \in S$, such a number m is called a *lower bound* for S .
- iii. *Bounded* if it is bounded above and below.

Factorials

Let n be a positive integer. By n *factorial*, denoted $n!$, we mean the product of all the natural numbers from n down to 1:

$$n! = n(n-1)(n-2)(n-3) \cdots 3 \cdot 2 \cdot 1$$

In particular

$$1! = 1,$$

$$2! = 2 \cdot 1 = 2$$

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$$

For convenience we define

$$0! = 1.$$

Powers and Roots

Let a a real number, p a positive integer :

$$a^1 = a,$$

$$a^p = \overbrace{a \cdot a \cdot a \cdots a}^{p \text{ factors}}$$

$$a \neq 0 : a^0 = 1, a^{-p} = 1/a^p$$

$$\text{laws of exponents} \quad a^{p+q} = a^p a^q, a^{pq} = a^p a^q, (a^q)^p = a^{pq}$$

a real, q odd $a^{1/q}$, called the q th root of a , is the number b such that $b^q = a$

a non-negative, q even $a^{1/q}$ is the non-negative number b such that $b^q = a$

notation $a^{1/q}$ can be written $\sqrt[q]{a}$ ($a^{1/2}$ is written $\sqrt{a}$)

rational exponents $a^{p/q} = (a^{1/q})^p$

Basic Formulas

$$\begin{aligned}
 (a+b)^2 &= a^2 + 2ab + b^2 \\
 (a-b)^2 &= a^2 - 2ab + b^2 \\
 (a+b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3 \\
 (a+b)^4 &= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4 \\
 (a-b)^3 &= a^3 - 3a^2b + 3ab^2 - b^3 \\
 a^2 - b^2 &= (a-b)(a+b) \\
 a^3 - b^3 &= (a-b)(a^2 + ab + b^2) \\
 a^3 + b^3 &= (a+b)(a^2 - ab + b^2)
 \end{aligned}$$

Recall **The Binomial expansion**,

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \text{ where } \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Quadratic Equations

The roots of a quadratic equation

$$ax^2 + bx + c = 0$$

are given by the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

If $b^2 - 4ac > 0$, the equation has two real roots, if $b^2 - 4ac = 0$, the equation has one real root; if $b^2 - 4ac < 0$, the equation has no real roots but it has two complex roots.

1.2 Functions

Definition 1 A **function** f is a rule that assigns to each element x in a set D exactly one element, called $f(x)$, in a set E .

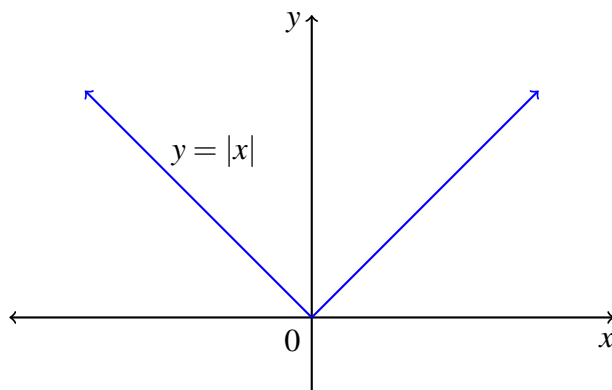
For our purposes, we shall consider functions for which the sets D and E are sets of real numbers. $y = f(x)$ is the **value of f at x** and is read "y equals f of x ". The set D of all possible input values is called the **domain** of the function. The **range** of f is the set of all output values of $f(x)$ as x varies throughout the domain D . A symbol that represents an arbitrary number in the domain of a function f is called an **independent variable**. A symbol that represents a number in the range is called a **dependent variable**. Functions are tools for describing the real world in Mathematical terms. A function can be represented by an equation, a graph, a numerical table, or verbal description.

1.2.1 Piecewise Defined Functions

Sometimes a function is defined by using different formulas on different parts of its domain. One such example is the **absolute value function**

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0, \end{cases}$$

whose graph is as shown.



The right-hand side of the equation means that the $y = x$ if $x \geq 0$ and the left-hand side means $y = -x$ if $x < 0$.

For example, $|5| = 5$, $|-5| = -(-5) = 5$, $|\pi - 5| = -(\pi - 5)$ (which is positive).

Geometric interpretation

- $|a|$ = distance between a and 0. (Distance is always positive or zero, so $|a| \geq 0$.)
- $|a - c|$ = distance between a and c .

The following characterizations are helpful:

Let a and c be two real constants, then

- $|a| = \max\{a, -a\}$.
- $|a| = \sqrt{a^2}$.
- $|x| \leq c$ implies that $-c \leq x \leq c$.
- $|x| \geq c$ implies that $x \leq -c$ or $x \geq c$.

Example 2 Defined $h(x)$ by

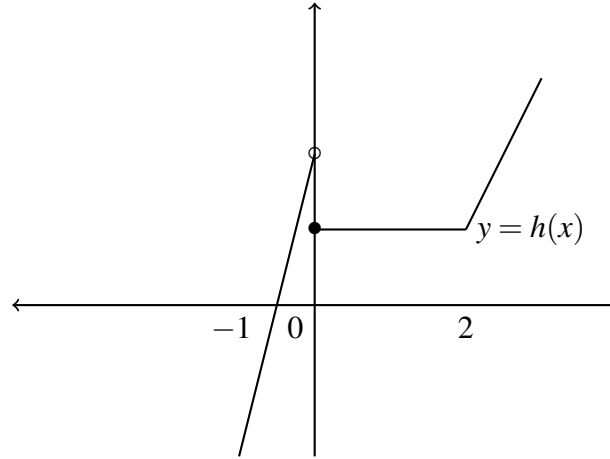
$$h(x) = \begin{cases} 2x + 1 & \text{if } x < 0 \\ 1 & \text{if } 0 \leq x < 2 \\ 2x - 3 & \text{if } x \geq 2, \end{cases}$$

Evaluate,

$h(-2), h(-1), h(0), h(1), h(2), h(5)$ and sketch the graph.

Solution 3

- $h(-2) = 2(-2) + 1 = -3$
- $h(-1) = 2(-1) + 1 = -1$
- $h(0) = 1$
- $h(1) = 1$
- $h(2) = 2(2) - 3 = 4 - 3 = 1$
- $h(5) = 2(5) - 3 = 10 - 3 = 7$



Example 4 A function f is defined by

$$f(x) = \begin{cases} 1-x, & x \leq -1 \\ x^2, & x > -1 \end{cases}$$

Evaluate $f(-2)$, $f(-1)$, and $f(0)$ and sketch the graph.

Example 5 Draw a sketch graph of the function given by

$$f(x) = \begin{cases} -x, & x < 0 \\ x^2, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases}$$

Solution 6 Exercise.

Example 7 The function whose value at any number x is the **greatest integer less than or equal to** x is called the **greatest integer function** or the **integer floor function**. It is denoted by $\lfloor x \rfloor$. Observe that

$$\begin{aligned} \lfloor 2.4 \rfloor &= 2 & \lfloor 1.9 \rfloor &= 1 \\ \lfloor 2 \rfloor &= 2 & \lfloor -0.3 \rfloor &= -1 \end{aligned}$$

Example 8 The function whose value at any number x is the **smallest integer greater than or equal to** x is called the **least integer function** or the **integer ceiling function**. It is denoted by $\lceil x \rceil$. Observe that

$$\begin{aligned} \lceil 2.4 \rceil &= 3 & \lceil 0.7 \rceil &= 1 \\ \lceil 1.8 \rceil &= 2 & \lceil -0.8 \rceil &= 0 \end{aligned}$$

1.2.2 Symmetry

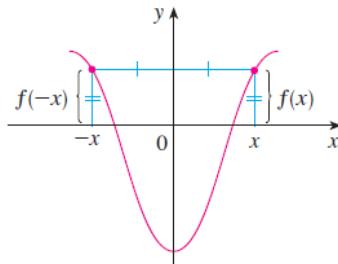
Definition 9 A function $y = f(x)$ is an

even function of x if $f(-x) = f(x)$

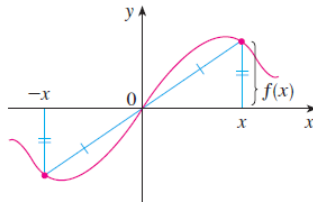
odd function of x if $f(-x) = -f(x)$

for every x in the function's domain.

The graph of an even function is symmetric about the y -axis. A reflection about the y -axis leaves the graph unchanged.



The graph of an odd function is symmetric about the origin. A rotation of the graph of f through π radians (180°) about the origin leaves the graph unchanged.



Example 10 Determine whether each of the following functions is even, odd or neither even or odd.

(a) $f(x) = x^7 + x$

(b) $g(x) = 1 - x^8$

(c) $h(x) = 4x - x^4$

Solution 11 Exercise.

1.2.3 Polynomials

A function P is called a polynomial if

$$\begin{aligned} P(x) &= a_n x^n + a_{n-1} x^{n-1} + \cdots + a_2 x^2 + a_1 x + a_0 \\ &= \sum_{k=0}^n a_k x^k \end{aligned}$$

where n is a positive integer and the numbers $a_0, a_1, \dots, a_n$ are constants called the coefficients of the polynomial. The domain of any polynomial is $\mathbb{R} = (-\infty, \infty)$. If the leading coefficient $a_n \neq 0$, then the **degree** of the polynomial is n .

1.2.4 Rational Functions

A rational function f is a ratio of two polynomials:

$$f(x) = \frac{P(x)}{Q(x)}$$

where P and Q are polynomials. The domain consist of all values of x such that $Q(x) \neq 0$.

1.2.5 Algebraic Functions

A function f is called an **algebraic function** if it can be constructed using algebraic operations (such as addition, subtraction, multiplication, division, and taking roots), starting with polynomials. Any rational function is automatically an algebraic function.

For example $g(x) = \frac{x^4 - 12x^2}{x + \sqrt{x}} + (x - 3)\sqrt[3]{x + 1}$.

1.2.6 Trigonometric Functions

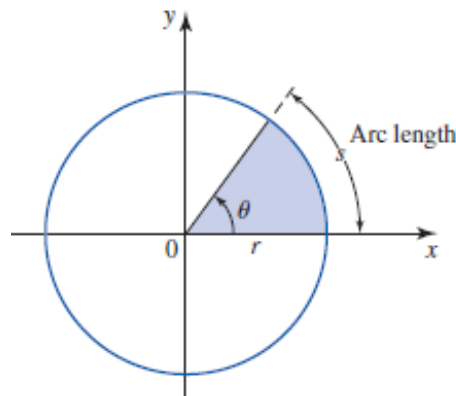
In calculus the convention is that **radian measure** is always used (except when indicated otherwise) to measure angles.

Definition 12 *Radian Measure of an Angle*

If s is the length of the arc subtended by a central angle θ in a circle of radius r , then

$$\theta = \frac{s}{r}$$

is the radian measure of θ .



Since the circumference of a unit circle is 2π and the central angle subtended by one complete revolution is 360° , we see that

$$2\pi \text{ radians (rad)} = 360^\circ$$

or

$$1 \text{ rad} = \left(\frac{180}{\pi} \right)^\circ$$

and

$$1^\circ = \frac{\pi}{180^\circ} \text{ rad or } \pi \text{ radians} = 180^\circ$$

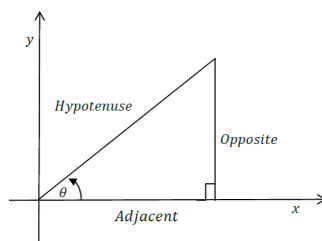
Converting degrees and radians

- To convert degrees to radians, multiply by $\frac{\pi}{180^\circ}$.
- To convert radians to degrees, multiply by $\frac{180^\circ}{\pi}$.

The table below shows the equivalence between degree and radian measures for some special angles.

Degrees	0°	30°	45°	60°	90°	120°	135°	150°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

We recall the six basic trigonometric ratios,



$$\begin{array}{ll}
 \text{sine:} & \sin \theta = \frac{\text{opp}}{\text{hyp}} \\
 \text{cosine:} & \cos \theta = \frac{\text{adj}}{\text{hyp}} \\
 \text{tangent:} & \tan \theta = \frac{\text{opp}}{\text{adj}} \\
 \text{cosecant:} & \csc \theta = \frac{\text{hyp}}{\text{opp}} \\
 \text{secant:} & \sec \theta = \frac{\text{hyp}}{\text{adj}} \\
 \text{cotangent:} & \cot \theta = \frac{\text{adj}}{\text{opp}}
 \end{array}$$

Notice that whenever the quotients are defined,

$$\begin{array}{ll}
 \tan \theta = \frac{\sin \theta}{\cos \theta} & \cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{1}{\tan \theta} \\
 \sec \theta = \frac{1}{\cos \theta} & \csc \theta = \frac{1}{\sin \theta}
 \end{array}$$

We list the following identities:

$\sin^2 x + \cos^2 x = 1$	$\cos^2 x = \frac{(1 + \cos 2x)}{2}$	$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
$\tan^2 x + 1 = \sec^2 x$	$\sin^2 x = \frac{(1 - \cos 2x)}{2}$	$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
$\cot^2 x + 1 = \csc^2 x$		
	Double angle	
$\sin 2x = 2 \sin x \cos x$	$\cos 2x = \cos^2 x - \sin^2 x$	

Length of a Circular Arc

$$s = r\theta$$

Area of a Circular Sector

$$A = \frac{1}{2}r^2\theta$$

1.2.7 Exponential Functions

Exponential functions are functions of the form

$$f(x) = a^x$$

where the base a is a positive constant. Revise the exponential laws.

1.2.8 Logarithmic Functions

The logarithmic functions

$$f(x) = \log_a x,$$

where the base a is a positive constant, are the inverse functions of the exponential functions. Later we shall discuss the **natural exponential and natural logarithmic functions** where the base is a special irrational number $e = 2.718181828\ldots$.

1.2.9 Combinations of Functions

Two functions f and g can be combined to form new functions in a manner similar to the way we add, subtract, multiply, and divide real numbers. The sum, difference, product and quotient functions are given by

$$\begin{aligned}(f+g)(x) &= f(x) + g(x) \\ (f-g)(x) &= f(x) - g(x) \\ (fg)(x) &= f(x)g(x) \\ \left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)}.\end{aligned}$$

Definition 13 Given two functions f and g , the **composite function** $f \circ g$ (called the composition of f and g) is defined by

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f .

Example 14 If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2-x}$, find each function and its domain.

$$(a) f \circ g \quad (b) g \circ f \quad (c) f \circ f \quad (d) g \circ g$$

Solution 15 (a) $(f \circ g)(x) = f(g(x)) = f(\sqrt{2-x}) = \sqrt{\sqrt{2-x}} = \sqrt[4]{2-x}$

Domain of $f \circ g$ is $\{x | 2-x \geq 0\} = \{x | x \leq 2\} = (-\infty, 2]$

(b) $(g \circ f)(x) = g(f(x)) = g(\sqrt{x}) = \sqrt{2-\sqrt{x}}$. For $\sqrt{x}$ to be defined we must have $x \geq 0$. For $\sqrt{2-\sqrt{x}}$ to be defined we must have $2-\sqrt{x} \geq 0$, that is $\sqrt{x} \leq 2$ or $x \leq 4$. Thus we have $0 \leq x \leq 4$, so the domain of $g \circ f$ is the closed interval $[0, 4]$. (c) and (d) are left as an exercise.

Exercise 16 .

1. Find the domain of the function

$$\begin{array}{lll}
 \text{(a)} \ f(x) = \frac{x+4}{x^2-9} & \text{(c)} \ h(x) = \frac{1}{\sqrt[4]{x^2-5x}} & \text{(e)} \ f(u) = \frac{u+1}{1+\frac{1}{u+1}} \\
 \text{(b)} \ g(t) = \sqrt[3]{2t-1} & \text{(d)} \ y = \sqrt{|x|} &
 \end{array}$$

2. Determine $f+g, f-g, f \cdot g, f/g$ and give the domain of each

$$\begin{array}{ll}
 \text{(a)} \ f(x) = 2x-3, g(x) = 2-x & \text{(c)} \ f(x) = \sqrt{x-1}, f(x) = x - \sqrt{x+1} \\
 \text{(b)} \ f(x) = x^2-1, g(x) = x+1/x & \text{(d)} \ f(x) = \sin^2 x, g(x) = \cos 2x
 \end{array}$$

3. Graph the following functions

$$\begin{array}{ll}
 \text{(a)} \ g(x) = \begin{cases} 1-x; & 0 \leq x \leq 1 \\ 2-x; & 1 < x \leq 2 \end{cases} & \text{(d)} \ f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2-x, & 1 < x \leq 2 \end{cases} \\
 \text{(b)} \ f(x) = \begin{cases} 2x+1; & x \leq 0 \\ x^2-x; & x > 0 \end{cases} & \text{(e)} \ f(x) = \begin{cases} \sqrt{-x}, & -4 \leq x \leq 0 \\ \sqrt{x}, & 0 < x \leq 4 \end{cases} \\
 \text{(c)} \ h(x) = \begin{cases} 1/x, & x < 0 \\ x, & 0 \leq x \end{cases} & \text{(f)} \ g(x) = \begin{cases} -x-1 & \text{if } x < -1 \\ 0 & \text{if } -1 \leq x \leq 1 \\ x+1 & \text{if } x > 1 \end{cases}
 \end{array}$$

4. Solve for x in the given interval $[0, 2\pi]$

$$\begin{array}{ll}
 \text{(a)} \ \sin x = 1/2 & \text{(c)} \ \cos x = -1/2 \\
 \text{(b)} \ \sin 2x = -\sqrt{3}/2 & \text{(d)} \ \tan(x/2) = 1
 \end{array}$$

5. Form the composition $f \circ g$ and give the domain

$$\begin{array}{ll}
 \text{(a)} \ f(x) = 2x+5, g(x) = x^2 & \text{(c)} \ f(x) = \sqrt{1-x^2}, g(x) = \cos 2x \\
 \text{(b)} \ f(x) = \sqrt{x}, g(x) = x^2+5 & \text{(d)} \ f(x) = 1/x, g(x) = (x-2)/x.
 \end{array}$$

6. Form the composition $f \circ g \circ h$

$$\begin{array}{ll}
 \text{(a)} \ f(x) = 4x, g(x) = x-1, h(x) = x^2 & \\
 \text{(b)} \ f(x) = x-1, g(x) = 4x, h(x) = x^2 & \\
 \text{(c)} \ f(x) = \frac{1}{x}, g(x) = \frac{1}{2x+1}, h(x) = x^2 & \\
 \text{(d)} \ f(x) = \frac{x+1}{x}, g(x) = \frac{1}{2x+1}, h(x) = x^2. &
 \end{array}$$

2 Limits

2.1 The Limit of a function

2.1.1 Two sided limits

The notion of a limit permeates much of our work in calculus. We begin with the intuitive introduction to the limits. We use limits to describe the way a function varies. Some functions vary continuously; small changes in x produce only small changes in $f(x)$.

Definition 17 Let f be a function defined on an open interval containing a , with the possible exception of a itself. Then the limit of $f(x)$ as x approaches a is the number L , written

$$\lim_{x \rightarrow a} f(x) = L$$

if $f(x)$ can be made arbitrarily close to L (as close as we like) by taking x to be sufficiently close to a (on either side of a) but not equal to a .

An alternative notation for

$$\lim_{x \rightarrow a} f(x) = L$$

is $f(x) \rightarrow L$ as $x \rightarrow a$.

Example 18 . Use the graph of the function f shown in the figure to find the given limit, if it exists,

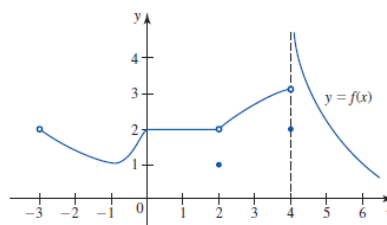
(a) $\lim_{x \rightarrow 1} f(x)$

(c) $\lim_{x \rightarrow 5} f(x)$

(e) $\lim_{x \rightarrow 10} f(x)$

(b) $\lim_{x \rightarrow 3} f(x)$

(d) $\lim_{x \rightarrow 7} f(x)$



Solution 19 (a) The values of f can be made as close to 2 as we please by taking x to be sufficiently close to 1. So $\lim_{x \rightarrow 1} f(x) = 2$

(b) Exercise

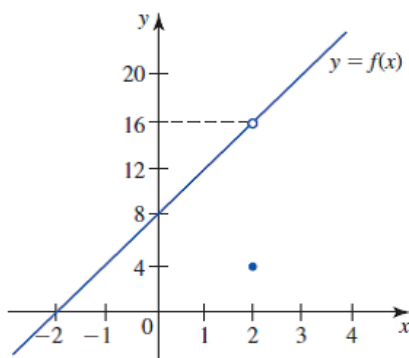
- (c) No matter how close x is to 5, there are values of f , corresponding to values of x smaller than 5, that are close to 1; and there are values of f , corresponding to values of x greater than 5, that are close to 4. In other words, there is no unique number that $f(x)$ approaches as x approaches 5. Therefore, $\lim_{x \rightarrow 5} f(x)$ does not exist. Observe that $f(5) = 1$, but this has no bearing on the existence or nonexistence of the limit.

(d) Exercise

(e) Exercise

Example 20 Find $\lim_{x \rightarrow 2} f(x)$ if it exists, where f is a piecewise-defined function

$$f(x) = \begin{cases} 4x + 8 & \text{if } x \neq 2 \\ 4 & \text{if } x = 2 \end{cases}$$



Solution 21 From the graph we see that $\lim_{x \rightarrow 2} f(x) = 16$.

Example 22 Draw a sketch graph and discuss the behaviour of the following functions, explaining why their limit as $x \rightarrow 0$ does not exist.

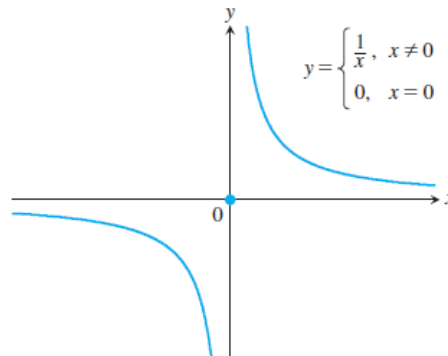
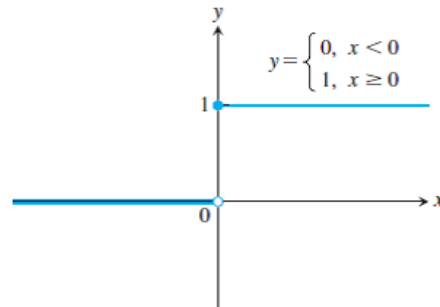
$$(a) \quad u(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

$$(b) \quad g(x) = \begin{cases} \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Solution 23 (a) The function jumps: The unit step function $u(x)$ has no limit as $x \rightarrow 0$ because its values jump at $x = 0$. For the negative values of x arbitrarily close to zero, $u(x) = 0$.

For positive values of x arbitrarily close to zero, $u(x) = 1$. There is no single value L approached by $u(x)$ as $x \rightarrow 0$.

(b) The function grows too "large" to have a limit: $g(x)$ has no limit as $x \rightarrow 0$ because the values of g grow arbitrarily large in absolute value as $x \rightarrow 0$ and therefore do not stay close to any fixed real number. We say the function is **not bounded**.



2.1.2 One-Sided Limits

Definition 24 Let f be a function defined for all values of x close to but greater than a . Then the right-hand limit of $f(x)$ as x approaches a is equal to L , written

$$\lim_{x \rightarrow a^+} f(x) = L$$

if $f(x)$ can be made as close to L as we please by taking x to be sufficiently close to but greater than a .

Definition 25 Let f be a function defined for all values of x close to but less than a . Then the left-hand limit of $f(x)$ as x approaches a is equal to L , written

$$\lim_{x \rightarrow a^-} f(x) = L$$

if $f(x)$ can be made as close to L as we please by taking x to be sufficiently close to but less than a .

The above two limits are often referred to as **one sided limits** whereas $\lim_{x \rightarrow a} f(x)$ is a **two sided limit**.

Let f be a function defined on an open interval containing a , with the possible exception of a itself. Then

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = L$$

Thus the (two sided) limit exists if and only if the one-sided limits exist and are equal.

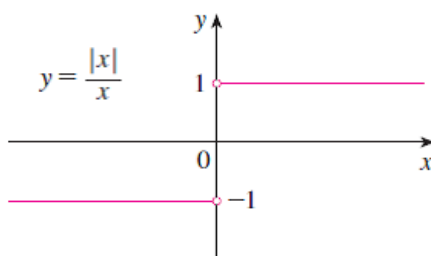
Example 26 Sketch the graph of f defined by

$$f(x) = \begin{cases} 3-x & \text{if } x < 1 \\ 1 & \text{if } x = 1 \\ 2 + \sqrt{x-1} & \text{if } x > 1 \end{cases}$$

Use the graph to find $\lim_{x \rightarrow 1^+} f(x)$, $\lim_{x \rightarrow 1^-} f(x)$ and $\lim_{x \rightarrow 1} f(x)$.

Solution 27 Exercise

Example 28 Find $\lim_{x \rightarrow 0} \frac{|x|}{x}$.



Solution 29

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{|x|}{x} &= \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1 \\ \lim_{x \rightarrow 0^-} \frac{|x|}{x} &= \lim_{x \rightarrow 0^-} \frac{-x}{x} = \lim_{x \rightarrow 0^-} -1 = -1 \end{aligned}$$

Since the right- and left-hand limits are different, it follows that $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

Example 30 If

$$f(x) = \begin{cases} \sqrt{x-4} & \text{if } x > 4 \\ 8-2x & \text{if } x < 4 \end{cases}$$

determine whether $\lim_{x \rightarrow 4} f(x)$ exists.

Solution 31 Exercise.

2.1.3 Calculating Limits Using the Limit Laws

Limit Laws Suppose that c is a constant and the limits

$$\lim_{x \rightarrow a} f(x) \quad \text{and} \quad \lim_{x \rightarrow a} g(x)$$

exist. Then

1. $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
2. $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$

3. $\lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x)$
4. $\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$
5. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$ if $\lim_{x \rightarrow a} g(x) \neq 0$

If we use the Product Law repeatedly with $g(x) = f(x)$, we obtain the following law.

$$6. \lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n \text{ where } n \text{ is a positive integer}$$

In applying these six limit laws, we need to use two special limits:

7. $\lim_{x \rightarrow a} c = c$
8. $\lim_{x \rightarrow a} x = a$

Direct Substitution Property: If f is a polynomial or a rational function and a is in the domain of f , then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Functions with Direct Substitution Property are called continuous at a and will be studied in more details later. However, not all limits can be evaluated by direct substitution.

Example 32 Find $\lim_{x \rightarrow \sqrt{2}} \left(\frac{x^2 - 2}{x - \sqrt{2}} \right)$

Solution 33 We factor the numerator as the difference of squares:

$$\begin{aligned} \lim_{x \rightarrow \sqrt{2}} \left(\frac{x^2 - 2}{x - \sqrt{2}} \right) &= \lim_{x \rightarrow \sqrt{2}} \frac{(x - \sqrt{2})(x + \sqrt{2})}{x - \sqrt{2}} \\ &= \lim_{x \rightarrow \sqrt{2}} (x + \sqrt{2}) = \sqrt{2} + \sqrt{2} = 2\sqrt{2} \end{aligned}$$

NOTE We computed the limit above by replacing the given function $f(x) = \frac{x^2 - \sqrt{2}}{x - \sqrt{2}}$

with a simpler function, $g(x) = x + \sqrt{2}$ which has the same limit as the original function. This is valid because $f(x) = g(x)$ except when $x = \sqrt{2}$, and in computing a limit as x approaches 1 we don't consider what happens when x is actually equal to $\sqrt{2}$. In general we have the following fact:

If $f(x) = g(x)$ when $x \neq a$, then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$, provided the limit exist.

Example 34 Find $\lim_{x \rightarrow 1} g(x)$ where

$$g(x) = \begin{cases} x + 1 & \text{if } x \neq 1 \\ \pi & \text{if } x = 1 \end{cases}$$

Solution 35 *Exercise.*

Example 36 *Evaluate*

$$\lim_{h \rightarrow 0} \frac{(3+h)^2 - 9}{h}$$

Solution 37 *Exercise*

Example 38 *Find*

$$\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 100} - 10}{x^2}$$

Solution 39 *Exercise*

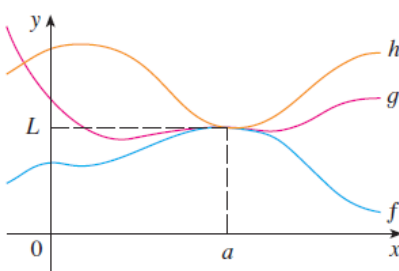
Theorem 40 (The Squeeze Theorem) If $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

then

$$\lim_{x \rightarrow a} g(x) = L$$

The Squeeze Theorem is sometimes called the Sandwich Theorem or the Pinching Theorem.



Example 41 *Show that*

$$\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0.$$

Solution 42 *Because the sine of any number lies between -1 and 1 , we can write*

$$-1 \leq \sin \frac{1}{x} \leq 1$$

Any inequality remains true when multiplied by a positive number. We know that $x^2 \geq 0$ for all x and so, multiplying each side of the inequality by x^2 , we get

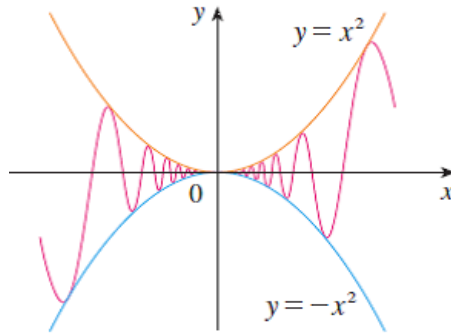
$$-x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$$

as illustrated in figure 2.7. We know that

$$\lim_{x \rightarrow 0} x^2 = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} (-x^2) = 0$$

Taking $f(x) = -x^2$, $g(x) = x^2 \sin \frac{1}{x}$, and $h(x) = x^2$ in the Squeeze Theorem, we obtain

$$\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$$

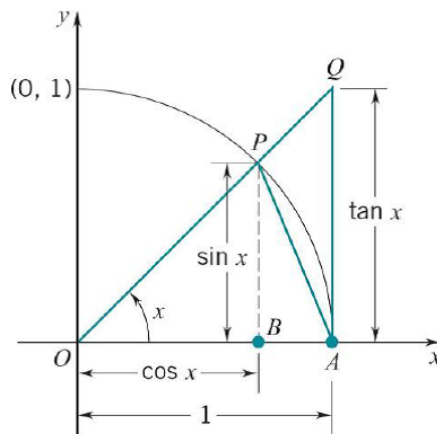


Theorem 43

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Proof.

First, suppose that $0 < x < \frac{\pi}{2}$. Figure 2.8 shows a sector with radius 1, length $PB = \sin x$, length $OB = \cos x$ and length $OA = 1$. Also OAQ is a right triangle, we have length



$$\begin{aligned} QA = \tan x. \text{ Now, } \quad \text{Area } \triangle OAP &= \frac{1}{2}(1) \sin x = \frac{\sin x}{2} && \text{Area sector} \\ OAP &= \frac{1}{2}(1)^2 x = \frac{x}{2} && \text{Area } \triangle OAQ = \frac{1}{2}(1) \tan x = \frac{\sin x}{2 \cos x}. \text{ Since} \\ &&& \triangle OAP \subseteq \text{sector } OAP \subseteq \triangle OAQ, \end{aligned}$$

we have ,

$$\begin{aligned}\frac{\sin x}{2} &< \frac{x}{2} < \frac{\sin x}{2\cos x} \\ 1 &< \frac{x}{\sin x} < \frac{1}{\cos x} \\ \cos x &< \frac{\sin x}{x} < 1\end{aligned}$$

We can apply Squeeze Theorem. Since

$$\lim_{x \rightarrow 0} \cos x = 1 \quad \text{and} \quad \lim_{x \rightarrow 0} 1 = 1$$

we can conclude that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

■

Now you can show that

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0.$$

Example 44 Find

$$\lim_{x \rightarrow 0} x \cot 3x$$

Example 45 Find

$$\lim_{x \rightarrow 0} \frac{\sin 2x}{5x}$$

Exercise 46 .

Evaluate the limit, if it exists

1. $\lim_{x \rightarrow -2} (x^2 - 1)$
2. $\lim_{x \rightarrow -2} (x^3 - 2x^2 + 4x + 8)$
3. $\lim_{s \rightarrow 2/3} (8 - 3s)(2s - 1)$
4. $\lim_{y \rightarrow 3} \frac{y + 3}{y^2 - 6y + 6}$
5. $\lim_{x \rightarrow 0} \left[\frac{1}{x\sqrt{1+x}} - \frac{1}{x} \right]$
6. $\lim_{x \rightarrow 9} \frac{x^2 - 81}{\sqrt{x} - 3}$
7. $\lim_{t \rightarrow 9} \frac{9 - t}{3 - \sqrt{t}}$
8. $\lim_{h \rightarrow 0} \frac{(3+h)^{-1} - 3^{-1}}{h}$
9. $\lim_{y \rightarrow 0} \frac{5y^3 + 8y^2}{3y^4 - 16y^2}$
10. $\lim_{x \rightarrow 3} \frac{2x - 6}{x - 3}$
11. $\lim_{x \rightarrow -3} (|x| - 2)$
12. $\lim_{x \rightarrow 2} (\sqrt{2x^3} - \sqrt{2x})$
13. $\lim_{y \rightarrow -3} (5 - y)^{4/3}$
14. $\lim_{x \rightarrow 0} \frac{\sec 2x}{\sqrt{x+4}}$
15. $\lim_{x \rightarrow 4} \sqrt{x^2 + 2x + 1}$
16. $\lim_{x \rightarrow 0} f(x); f(x) = \begin{cases} 1, & x \neq 0 \\ 3, & x = 0 \end{cases}$
17. $\lim_{x \rightarrow 1} f(x); f(x) = \begin{cases} 3x, & x < 1 \\ x + 2, & x \geq 1 \end{cases}$

$$18. \lim_{x \rightarrow 1} f(x); f(x) = \begin{cases} 3x, & x < 1 \\ 3, & x > 1 \end{cases}$$

$$21. \lim_{x \rightarrow 0} \frac{x}{(x+2)^2 - (x-2)^2}$$

$$19. \lim_{x \rightarrow 0} f(x); f(x) = \begin{cases} 1, & x \neq 0 \\ 3, & x = 0 \end{cases}$$

$$22. \lim_{x \rightarrow 0} \frac{\frac{1}{x-1} + \frac{1}{x+1}}{x}$$

$$20. \lim_{w \rightarrow 0} \frac{\sqrt{w+1} - \sqrt{w^2+4}}{(w+2)^2 - (w+1)^2}$$

$$23. \lim_{x \rightarrow -1} \frac{\sqrt{x^2+8} - 3}{x+1}$$

$$24. \lim_{x \rightarrow 0} \sin^2 x$$

$$26. \lim_{x \rightarrow 2} \frac{1+x+\sin x}{3\cos x}$$

$$28. \lim_{x \rightarrow 4} \frac{4-x}{5-\sqrt{x^2+9}}$$

$$25. \lim_{x \rightarrow 0} \sec x$$

$$27. \lim_{x \rightarrow 0} (2\sin x - 1)$$

Using $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$, find the following limits:

$$29. \lim_{\theta \rightarrow 0} \frac{\sin \sqrt{3\theta}}{\sqrt{3\theta}}$$

$$36. \lim_{x \rightarrow 0} \frac{\sin^2 3x}{5x^2}$$

$$43. \lim_{x \rightarrow \pi/4} \frac{1 - \cos x}{x}$$

$$30. \lim_{t \rightarrow 0} \frac{\sin 3t}{\sin 2t}$$

$$37. \lim_{t \rightarrow 0} \frac{\sin kt}{t}$$

$$44. \lim_{x \rightarrow 0} \frac{x^2 - 2x}{\sin 3x}$$

$$31. \lim_{y \rightarrow 0} \frac{\sin 3y}{4y}$$

$$38. \lim_{x \rightarrow 0} \frac{1 - \sec^2 2x}{x^2}$$

$$45. \lim_{x \rightarrow 0} \frac{2x^2 + x}{\sin x}$$

$$32. \lim_{h \rightarrow 0^-} \frac{h}{\sin 3h}$$

$$39. \lim_{x \rightarrow 0} \frac{\sin^2 x^2}{x^2}$$

$$46. \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\sin 2\theta}$$

$$33. \lim_{x \rightarrow 0} \frac{\tan 2x}{x}$$

$$40. \lim_{x \rightarrow 1} \frac{\sin(x-1)}{x^2 + x - 2}$$

$$47. \lim_{x \rightarrow 0} \frac{\sec x - 1}{x \sec x}$$

$$34. \lim_{x \rightarrow 0} \frac{\sin 4x}{\sin 6x}$$

$$41. \lim_{x \rightarrow 0} \frac{\sin^2 x}{x(1 - \cos^2 x)}$$

$$48. \lim_{\theta \rightarrow 0} \frac{\sin(\sin \theta)}{\sin \theta}$$

$$35. \lim_{x \rightarrow 0} \frac{\sin 3x \sin 5x}{x^2}$$

$$42. \lim_{x \rightarrow 0} \frac{\cos x - 1}{2x}$$

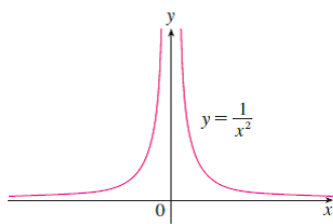
2.2 Infinite Limits

Example 47 Find $\lim_{x \rightarrow 0} \frac{1}{x^2}$ if it exists.

Solution 48 As x becomes close to 0, x^2 also becomes close to 0, and $\frac{1}{x^2}$ becomes very large. In fact from the graph of $f(x) = 1/x^2$ the values of $f(x)$ can be made arbitrarily large by taking x close enough to 0. The value of $f(x)$ does not approach a number so $\lim_{x \rightarrow 0} \frac{1}{x^2}$ does not exist and as such we write

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty.$$

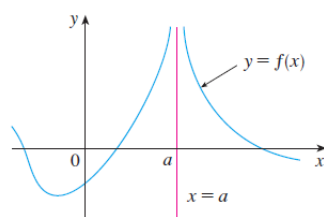
Note that ∞ is not a number.



Definition 49 Let f be a function defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = \infty.$$

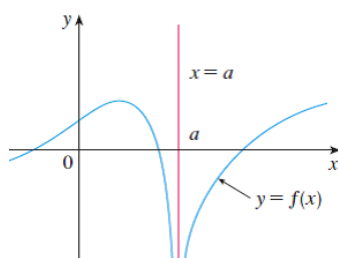
means that the values of $f(x)$ can be made arbitrarily large (as large as we please) by taking x sufficiently close to a , but not equal to a .



Definition 50 Let f be a function defined on both sides of a , except possibly at a itself. Then

$$\lim_{x \rightarrow a} f(x) = -\infty.$$

means that the values of $f(x)$ can be made arbitrarily large negative (as large as we please) by taking x sufficiently close to a , but not equal to a .



Similar definitions can be given for one sided infinite limits

$$\begin{array}{ll} \lim_{x \rightarrow a^-} f(x) = \infty. & \lim_{x \rightarrow a^+} f(x) = \infty. \\ \lim_{x \rightarrow a^-} f(x) = -\infty. & \lim_{x \rightarrow a^+} f(x) = -\infty. \end{array}$$

2.2.1 Vertical and horizontal asymptotes

Definition 51 The line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true:

$$\begin{array}{lll} \lim_{x \rightarrow a^-} f(x) = \infty & \lim_{x \rightarrow a^-} f(x) = \infty. & \lim_{x \rightarrow a^+} f(x) = \infty. \\ \lim_{x \rightarrow a^-} f(x) = -\infty & \lim_{x \rightarrow a^-} f(x) = -\infty. & \lim_{x \rightarrow a^+} f(x) = -\infty. \end{array}$$

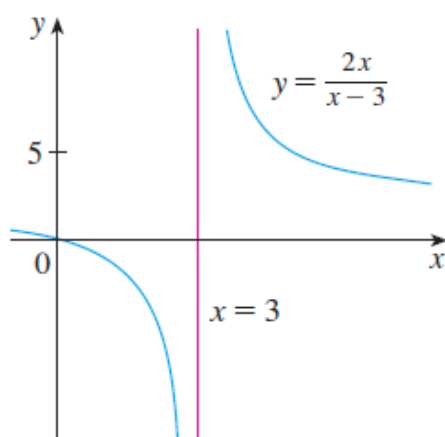
Example 52 Find $\lim_{x \rightarrow 3^+} \frac{2x}{x-3}$ and $\lim_{x \rightarrow 3^-} \frac{2x}{x-3}$.

Solution 53 If x is close to 3 but larger than 3, then the denominator $x-3$ is a small positive number and $2x$ is close to 6. So the quotient is a large positive number. Thus

$$\lim_{x \rightarrow 3^+} \frac{2x}{x-3} = \infty$$

Likewise if x is close to 3 but smaller than 3, then $x-3$ is a small negative number but $2x$ is still a positive number. So $\frac{2x}{x-3}$ is a numerically large negative number. Thus.

$$\lim_{x \rightarrow 3^-} \frac{2x}{x-3} = -\infty$$



This shows that the line $x = 3$ is a vertical asymptote of the curve $y = \frac{2x}{x-3}$.

Definition 54 Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large.

The geometric illustrations of the above definition are shown in the figure below. Notice that there are many ways for the graph of f to approach the line $y = L$ (which is called the **horizontal asymptote**) as we look to the far right of each graph.

The general definition is as follows. Let f be a function defined on some interval $(-\infty, a)$. Then

$$\lim_{x \rightarrow -\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large negative.

Theorem 55 If $r > 0$ is a rational number, then

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$$

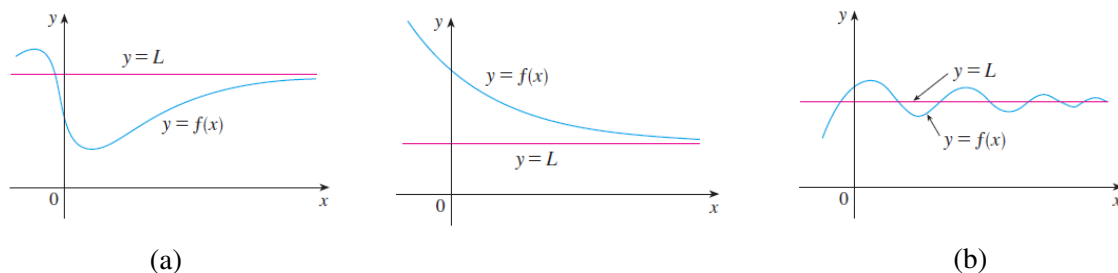
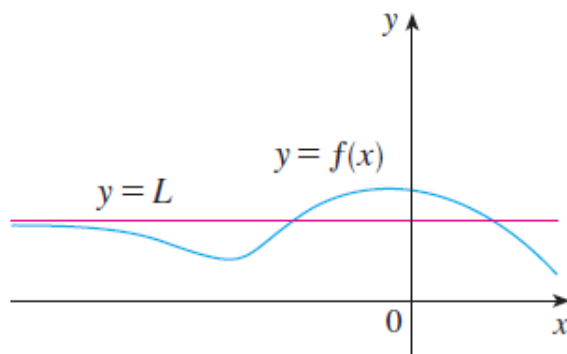


Figure 2.1: Horizontal asymptotes



If $r > 0$ is a rational number such that x^r is defined for all x , then

$$\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$$

Example 56 Evaluate

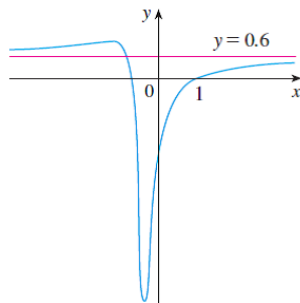
$$\lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 3x + 1}$$

and indicate which properties of limits are used at each stage.

Solution 57 As x becomes large, both numerator and denominator become large, so it isn't obvious what happens to the ratio. We need to do some preliminary algebra. To evaluate the limit at infinity of any rational function, we first divide both the numerator and denominator by the highest power of x that occurs in the denominator. (We may assume that $x \neq 0$, since we are interested only in large values of x .) In this case the highest power of x in the denominator is x^2 , so we have

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 3x + 1} &= \lim_{x \rightarrow \infty} \frac{\frac{3x^2 - x - 2}{x^2}}{\frac{5x^2 + 3x + 1}{x^2}} = \lim_{x \rightarrow \infty} \frac{3 - \frac{1}{x} - \frac{2}{x^2}}{5 + \frac{3}{x} + \frac{1}{x^2}} \\ &= \frac{\lim_{x \rightarrow \infty} \left(3 - \frac{1}{x} - \frac{2}{x^2} \right)}{\lim_{x \rightarrow \infty} \left(5 + \frac{3}{x} + \frac{1}{x^2} \right)} = \frac{\lim_{x \rightarrow \infty} \left(3 - \frac{1}{x} - \frac{2}{x^2} \right)}{\lim_{x \rightarrow \infty} \left(5 + \frac{3}{x} + \frac{1}{x^2} \right)} \\ &= \frac{\lim_{x \rightarrow \infty} 3 - \lim_{x \rightarrow \infty} \frac{1}{x} - \lim_{x \rightarrow \infty} \frac{2}{x^2}}{\lim_{x \rightarrow \infty} 5 + \lim_{x \rightarrow \infty} \frac{3}{x} + \lim_{x \rightarrow \infty} \frac{1}{x^2}} = \frac{3 - 0 - 0}{5 + 0 + 0} = \frac{3}{5} \end{aligned}$$

A similar calculation shows that the limit as $x \rightarrow -\infty$ is also $\frac{3}{5}$. The figure below illustrates the results of these calculations by showing how the graph of the given rational function approaches the horizontal asymptote $y = \frac{3}{5}$.



Definition 58 A line $y = b$ is a horizontal asymptote of the graph of a function $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = b \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = b.$$

Definition 59 A line $x = a$ is a vertical asymptote of the graph of a function $y = f(x)$ if either

$$\lim_{x \rightarrow a^+} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \rightarrow a^-} f(x) = \pm\infty.$$

Example 60 Find the horizontal and vertical asymptotes of the graph of

$$f(x) = -\frac{8}{x^2 - 4}$$

Solution 61 We are interested in the behaviour as $x \rightarrow \pm\infty$ and as $x \rightarrow \pm 2$, where the denominator is zero. Notice that f is an even function of x , so its graph is symmetric with respect to the y -axis.

(a) The behaviour as $x \rightarrow \pm\infty$. Since $\lim_{x \rightarrow \infty} f(x) = 0$, the line $y = 0$ is a horizontal asymptote of the graph to the right. By symmetry it is an asymptote to the left as well. (see the figure below).

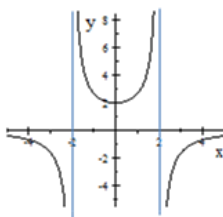
(b) The behaviour as $x \rightarrow \pm 2$. Since

$$\lim_{x \rightarrow 2^+} f(x) = -\infty \quad \text{and} \quad \lim_{x \rightarrow 2^-} f(x) = \infty$$

the line $x = 2$ is the vertical asymptote both from the right and from the left. By symmetry, the line $x = -2$ is also a vertical asymptote.

Example 62 Find the horizontal and vertical asymptotes of the graph of the function

$$f(x) = \frac{\sqrt{2x^2 + 1}}{3x - 5}$$



Solution 63 Dividing both the numerator and the denominator by x and using the properties of limits, we have

$$\lim_{x \rightarrow \infty} \frac{\sqrt{2x^2 + 1}}{3x - 5} = \lim_{x \rightarrow \infty} \frac{\sqrt{2 + \frac{1}{x^2}}}{3 - \frac{5}{x}} = \frac{\sqrt{2 + 0}}{3 - 5 \cdot 0} = \frac{\sqrt{2}}{3}.$$

Therefore the line $y = \sqrt{2}/3$ is a horizontal asymptote of the graph of f . In computing the limit as $x \rightarrow -\infty$, we must remember that for $x < 0$, we have $\sqrt{x^2} = |x| = -x$. So when we divide the numerator by x , for $x < 0$ we get

$$\frac{1}{x} \sqrt{2x^2 + 1} = -\frac{1}{\sqrt{x^2}} \sqrt{2x^2 + 1} = -\sqrt{2 + \frac{1}{x^2}}$$

Therefore

$$\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2 + 1}}{3x - 5} = \lim_{x \rightarrow -\infty} \frac{-\sqrt{2 + \frac{1}{x^2}}}{3 - \frac{5}{x}} = -\frac{\sqrt{2}}{3}$$

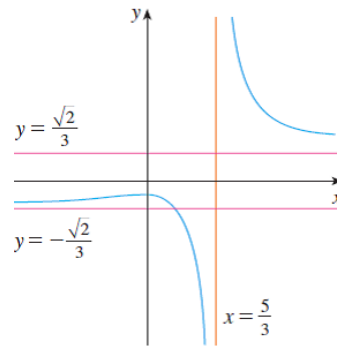
Thus the line $y = -\frac{\sqrt{2}}{3}$ is also the horizontal asymptote. A vertical asymptote is likely to occur when the denominator $3x - 5$, is 0, that is when $x = \frac{5}{3}$. If x is close to $\frac{5}{3}$ and $x > \frac{5}{3}$, then the denominator is close to 0 and $3x - 5$ is positive. The numerator $\sqrt{2x^2 + 1}$ is always positive, so $f(x)$ is positive. Therefore

$$\lim_{x \rightarrow (5/3)^+} \frac{\sqrt{2x^2 + 1}}{3x - 5} = \infty$$

If x is close to $\frac{5}{3}$ but $x < \frac{5}{3}$, then $3x - 5 < 0$ and so $f(x)$ is large negative. Thus

$$\lim_{x \rightarrow (5/3)^-} \frac{\sqrt{2x^2 + 1}}{3x - 5} = -\infty$$

The vertical asymptote is $x = \frac{5}{3}$. All three asymptotes are shown in the figure below.



Example 64 Compute

$$\lim_{x \rightarrow -\infty} \sqrt{x^2 + 2x + 1} + x$$

Solution 65 We first multiply numerator and denominator by conjugate radical:

$$\begin{aligned} \lim_{x \rightarrow -\infty} \sqrt{x^2 + 2x + 1} + x &= \lim_{x \rightarrow -\infty} \left(\sqrt{x^2 + 2x + 1} + x \right) \frac{\sqrt{x^2 + 2x + 1} - x}{\sqrt{x^2 + 2x + 1} - x} \\ &= \lim_{x \rightarrow -\infty} \frac{x^2 + 2x + 1 - x^2}{\sqrt{x^2 + 2x + 1} - x} \\ &= \lim_{x \rightarrow -\infty} \frac{2x + 1}{\sqrt{x^2 + 2x + 1} - x} \\ &\quad (\text{since } x \rightarrow -\infty, \text{ we must remember that for } x < 0, \text{ we have } \sqrt{x^2} = |x| = -x). \\ &= \lim_{x \rightarrow -\infty} \frac{-2 - \frac{1}{x}}{\sqrt{1 + \frac{2}{x} + \frac{1}{x^2}} + 1} = -1 \end{aligned}$$

2.2.2 Oblique Asymptotes

If the degree of the numerator of a rational function is 1 greater than the degree of the denominator, the graph has an oblique or slant line asymptote. We find an equation for the asymptote by dividing numerator by denominator to express f as a linear function plus a remainder that goes to zero as $x \rightarrow \pm\infty$.

Example 66 Find the oblique asymptote of the graph of

$$f(x) = \frac{x^2 - 3}{2x - 4}$$

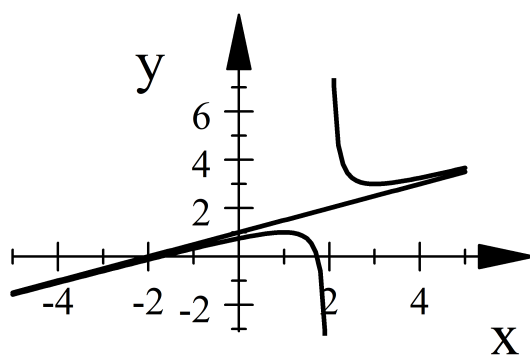
Solution 67 We are interested in the behaviour as $x \rightarrow \pm\infty$. We divide $(x^2 - 3)$ into $(2x - 4)$ using long division to get

$$f(x) = \frac{x^2 - 3}{2x - 4} = \underbrace{\left(\frac{x}{2} + 1 \right)}_{\text{linear } g(x)} + \underbrace{\left(\frac{1}{2x - 4} \right)}_{\text{remainder}}$$

As $x \rightarrow \pm\infty$, the remainder, whose magnitude gives the vertical distance between the graphs of f and g , goes to zero making the slanted line

$$g(x) = \frac{x}{2} + 1$$

An asymptote of the graph of f as shown below.



Exercise 68 .

Find the limit if it exists

1. $\lim_{x \rightarrow \infty} \sqrt{\frac{8x^2 - 3}{2x^2 + x}}$

8. $\lim_{x \rightarrow \infty} \frac{x^{-1} + x^{-4}}{x^{-2} - x^{-3}}$

2. $\lim_{x \rightarrow \infty} \left(\frac{x^2 + x - 1}{8x^2 - 3} \right)^{1/3}$

9. $\lim_{x \rightarrow \infty} (\sqrt{x+9} - \sqrt{x+4})$

3. $\lim_{x \rightarrow \infty} \frac{2\sqrt{x} + x^{-1}}{3x - 7}$

10. $\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 3} + x)$

4. $\lim_{x \rightarrow \infty} \frac{\sqrt[3]{x} - 5x + 3}{2x + x^{2/3} - 4}$

11. $\lim_{x \rightarrow \infty} (\sqrt{x^2 + 25} - \sqrt{x^2 - 1})$

5. $\lim_{x \rightarrow \infty} \frac{x - 3}{\sqrt{4x^2 + 25}}$

12. $\lim_{x \rightarrow \infty} (\sqrt{9x^2 - x} - 3x)$

6. $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 1}}{x + 1}$

13. $\lim_{x \rightarrow -\infty} (2x + \sqrt{4x^2 + 3x - 2})$

7. $\lim_{x \rightarrow \infty} \frac{\sqrt[3]{x} - \sqrt[5]{x}}{\sqrt[3]{x} + \sqrt[5]{x}}$

14. $\lim_{x \rightarrow -\infty} (\sqrt{x^2 + 3x} - \sqrt{x^2 - 2x})$

2.3 The Precise Definition of the Limit

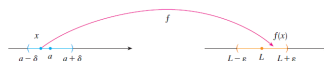
Definition 69 Let f be a function defined on some open interval that contains the number a , except possibly at a itself. Then we say that the limit of $f(x)$ as x approaches a is L , and we write

$$\lim_{x \rightarrow a} f(x) = L$$

if for every number $\varepsilon > 0$, we can find a number $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \text{ whenever } 0 < |x - a| < \delta$$

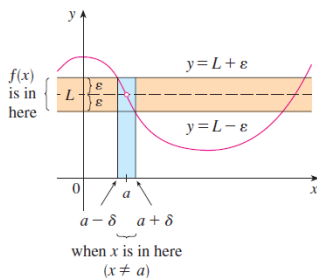
The definition of the limits says that if any small interval $(L - \varepsilon, L + \varepsilon)$ is given around L , then we can find an interval $(a - \delta, a + \delta)$ around a such that f maps all points in $(a - \delta, a + \delta)$ into the interval $(L - \varepsilon, L + \varepsilon)$. Another geometric interpretation of limits



can be given in terms of the graph of a function. If $\varepsilon > 0$ is given, then we can draw horizontal lines $y = L + \varepsilon$ and $y = L - \varepsilon$ and the graph of f . If $\lim_{x \rightarrow a} f(x) = L$, then we can find a number $\delta > 0$ such that if we restrict x to lie in the interval $(a - \delta, a + \delta)$ and take $x \neq a$ then the curve $y = f(x)$ lie between the lines $y = L - \varepsilon$ and $y = L + \varepsilon$.

Example 70 Prove that $\lim_{x \rightarrow 3} (3x - 5) = 4$.

Given $\varepsilon > 0$, we seek $\delta > 0$ such that $|(3x - 5) - 4| < \varepsilon$ whenever $0 < |x - 3| < \delta$.



Finding δ :

$$\text{Now } |(3x - 5) - 4| = |3x - 9| = 3|x - 3| < 3\delta.$$

That is $|x - 3| < \frac{\epsilon}{3}$ whenever $0 < |x - 3| < \delta$. This suggests we should choose $\delta = \frac{\epsilon}{3}$.

Proof: Given $\epsilon > 0$, choose $\delta = \frac{\epsilon}{3}$. Suppose $0 < |x - 3| < \delta$ then

$$|(3x - 5) - 4| = |3x - 9| = 3|x - 3| < 3\delta = 3\left(\frac{\epsilon}{3}\right) = \epsilon. \text{ Thus } |(3x - 5) - 4| < \epsilon$$

whenever $0 < |x - 3| < \delta$. Therefore by definition of the limit,

$$\lim_{x \rightarrow 3} (3x - 5) = 4.$$

Example 71 Prove that

$$\lim_{x \rightarrow 5} x^2 = 25$$

Finding δ :

Let $\epsilon > 0$ be given. We seek a $\delta > 0$ such that

$$|x^2 - 25| < \epsilon \text{ whenever } 0 < |x - 5| < \delta.$$

But

$$\begin{aligned} |x^2 - 25| &= |(x + 5)(x - 5)| \\ &= |x + 5| |x - 5| \end{aligned}$$

Notice that if we can find a positive constant C such that $|x + 5| < C$, then

$$|x + 5| |x - 5| < C |x - 5|$$

To find C , we restrict x to lie in some interval centered at 5. It is reasonable to assume that x lies within a distance 1 from 5, that is ,

$$|x - 5| < 1$$

then

$$\begin{aligned} -1 &< x - 5 < 1 \\ 4 &< x < 6 \end{aligned}$$

so

$$9 < x + 5 < 11$$

Thus we have

$$|x + 5| < 11$$

and so $C = 11$ is the reasonable choice. But there are two restrictions on $|x - 5|$, namely

$$|x - 5| < 1 \text{ and } |x - 5| < \frac{\varepsilon}{C} = \frac{\varepsilon}{11}$$

So we choose

$$\delta = \min \left\{ 1, \frac{\varepsilon}{11} \right\}$$

Proof:

Given $\varepsilon > 0$, choose $\delta = \min \left\{ 1, \frac{\varepsilon}{11} \right\}$. Suppose $0 < |x - 5| < \delta$, then

$$\begin{aligned} |x - 5| < 1 &\implies -1 < x - 5 < 1 \\ 4 < x < 6 &\implies |x + 5| < 11 \end{aligned}$$

We also have

$$|x - 5| < \frac{\varepsilon}{11}$$

so

$$|x^2 - 25| = |x + 5| |x - 5| < 11 \cdot \frac{\varepsilon}{11} = \varepsilon$$

This shows that

$$\lim_{x \rightarrow 5} x^2 = 25.$$

Alternative Method of finding δ

Given $\varepsilon > 0$, we seek a $\delta > 0$ such that

$$|x^2 - 25| < \varepsilon \text{ whenever } 0 < |x - 5| < \delta$$

We note that whenever

$$0 < |x - 5| < \lambda < 1, \text{ then } |x - 5|^n < \lambda \text{ for } n \in \mathbb{N}$$

Then

$$\begin{aligned} |x^2 - 25| &= |(x - 5)^2 + 10x - 50| \\ &= |(x - 5)^2 + 10(x - 5)| \\ &\leq |x - 5|^2 + 10|x - 5| \text{ (Triangle Inequality)} \\ &< \lambda + 10\lambda = 11\lambda \end{aligned}$$

Now $11\lambda < \varepsilon$ requires $\lambda < \frac{\varepsilon}{11}$.

Thus we choose $\delta = \min \left\{ 1, \frac{\varepsilon}{11} \right\}$

Proof:

Given $\varepsilon > 0$, choose $\delta = \min \left\{ 1, \frac{\varepsilon}{11} \right\}$. Suppose $0 < |x - 5| < \delta$, then

$$\begin{aligned} |x - 5| < 1 &\implies -1 < x - 5 < 1 \\ 4 < x < 6 &\implies |x + 5| < 11 \end{aligned}$$

We also have

$$|x - 5| < \frac{\varepsilon}{11}$$

so

$$|x^2 - 25| = |x + 5| |x - 5| < 11 \cdot \frac{\varepsilon}{11} = \varepsilon$$

This shows that

$$\lim_{x \rightarrow 5} x^2 = 25.$$

and the limit is established.

Example 72 Show that

$$\lim_{x \rightarrow 25} \sqrt{x} = 5$$

Solution 73 Given $\varepsilon > 0$, we seek $\delta > 0$ such that

$$|\sqrt{x} - 5| < \varepsilon \text{ whenever } 0 < |x - 25| < \delta.$$

Finding δ :

$$\begin{aligned} |\sqrt{x} - 5| &= \left| (\sqrt{x} - 5) \cdot \frac{(\sqrt{x} + 5)}{(\sqrt{x} + 5)} \right| \\ &= \left| \frac{x - 25}{\sqrt{x} + 5} \right| \text{ Note that } \sqrt{x} + 5 \text{ is greater than } 5, \text{ so} \\ &< \frac{|x - 25|}{5} < \varepsilon \end{aligned}$$

Thus $|x - 25| < 5\varepsilon$ and so $\delta = 5\varepsilon$.

Proof: Given $\varepsilon > 0$. Choose $\delta = 5\varepsilon$. Suppose $0 < |x - 25| < \delta$, then

$$\begin{aligned} |\sqrt{x} - 5| &= \left| (\sqrt{x} - 5) \cdot \frac{(\sqrt{x} + 5)}{(\sqrt{x} + 5)} \right| \\ &= \left| \frac{x - 25}{\sqrt{x} + 5} \right| \\ &< \frac{1}{5} |x - 25| < \frac{1}{5} \cdot 5\varepsilon = \varepsilon \end{aligned}$$

Thus $\lim_{x \rightarrow 25} \sqrt{x} = 5$.

Exercise 74 Use the precise definition of the limit to prove the following :

$$1. \lim_{x \rightarrow 2} 3x = 6$$

$$5. \lim_{x \rightarrow 2} (x^2 - 3x + 5) = 3$$

$$2. \lim_{x \rightarrow -2} (2x + 3) = -1$$

$$6. \lim_{x \rightarrow 4} \sqrt{x} = 2$$

$$3. \lim_{x \rightarrow 2} x^2 = 4$$

$$7. \lim_{x \rightarrow 3} \sqrt{3x - 5} = 2$$

$$4. \lim_{x \rightarrow 1} (2x^2 - 7) = -5$$

$$8. \lim_{x \rightarrow 2} x^3 = 8$$

2.4 Continuity

Previously, we noticed that the limit of a function as x approaches a can often be found simply by calculating the value of a function at a . Functions with this property are called *continuous functions*.

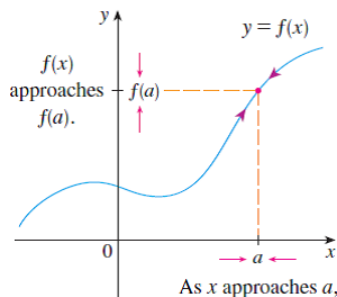
Definition 75 A function f is continuous at a if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

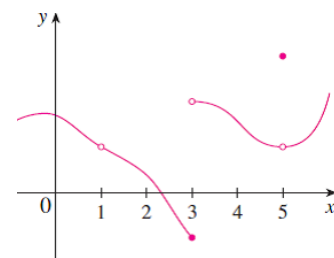
Notice that the definition implicitly requires three things if f is continuous at a :

1. $f(a)$ is defined (that is, a is in the domain of f)
2. $\lim_{x \rightarrow a} f(x)$ exists
3. $\lim_{x \rightarrow a} f(x) = f(a)$

A continuous function has the property that a small change in x produces only a small change in $f(x)$. If f is defined near a (in other words, f is defined on an open interval containing a , except perhaps at a) we say that f is **discontinuous at a** (or f **has a discontinuity at a**) if f is not continuous at a .



(a)



(b)

Example 76 Figure 2.21 (b) shows the graph of a function f . At which numbers is f discontinuous? Why?

Solution 77 It looks as if there is a discontinuity when $a = 1$ because the graph has a break there. The official reason that f is discontinuous at 1 is that $f(1)$ is not defined. The graph also has a break at $a = 3$, but the reason for discontinuity is different. Here $f(3)$ is defined, but $\lim_{x \rightarrow 3} f(x)$ does not exist because the left and right limits are different. So f is discontinuous at 3. What about $a = 5$? Here, $f(5)$ is defined and $\lim_{x \rightarrow 5} f(x)$ exists (because the left and right limits are the same). But

$$\lim_{x \rightarrow 5} f(x) \neq f(5)$$

So f is discontinuous at 5.

Example 78 Where are each of the following functions discontinuous?

$$(a) f(x) = \frac{x^2 - x - 2}{x - 2}$$

$$(b) f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

$$(c) f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

Solution 79 .

(a) Notice that $f(2)$ is not defined, so f is discontinuous at 2.

(b) Here $f(0) = 1$ is defined but

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{1}{x^2}$$

does not exist. So f is discontinuous at 0.

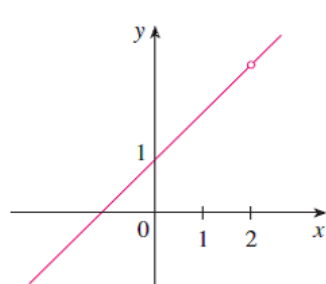
(c) Here $f(2) = 1$ is defined and

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 1)}{x - 2} = \lim_{x \rightarrow 2} (x + 1) = 3$$

exists. But

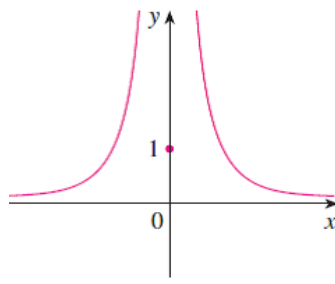
$$\lim_{x \rightarrow 2} f(x) \neq f(2)$$

The figure below shows the graphs of the functions in the above example. In each case the graph can't be drawn without lifting the pen from the paper because a hole or break or jump occurs in the graph. The kind of discontinuity illustrated in parts (a) and (c) is called removable because we could remove the discontinuity by redefining f at just the single number 2. [The function $g(x) = x + 1$ is continuous]. The discontinuity in part (b) is called an infinite discontinuity.



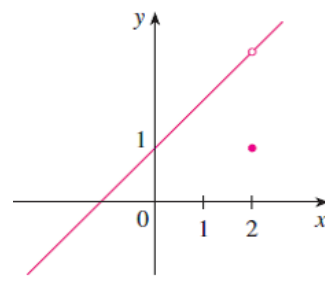
$$(a) f(x) = \frac{x^2 - x - 2}{x - 2}$$

(a)



$$(b) f(x) = \begin{cases} \frac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

(b)



$$(c) f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$$

(c)

Definition 80 A function $y = f(x)$ is **continuous at an interior point** a of its domain if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

A function $y = f(x)$ is **continuous at a left endpoint** a or **continuous at a right endpoint** b of its domain if

$$\lim_{x \rightarrow a^+} f(x) = f(a) \quad \text{or} \quad \lim_{x \rightarrow b^-} f(x) = f(b), \text{ respectively.}$$

Definition 81 A function f is **continuous on an interval** if it is continuous at every number in the interval. (If f is defined only on one side of the endpoint of the interval, we understand continuous at the endpoint to mean continuous from the right or continuous from the left.)

Example 82 Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.

Solution 83 If $-1 < a < 1$, then using the limit laws, we have

$$\begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} (1 - \sqrt{1 - x^2}) \\ &= 1 - \lim_{x \rightarrow a} \sqrt{1 - x^2} \\ &= 1 - \sqrt{\lim_{x \rightarrow a} (1 - x^2)} \\ &= 1 - \sqrt{1 - a^2} = f(a) \end{aligned}$$

Thus f is continuous at a if $-1 < a < 1$. Similar calculations show that

$$\lim_{x \rightarrow -1^+} f(x) = 1 = f(-1) \quad \text{and} \quad \lim_{x \rightarrow 1^-} f(x) = 1 = f(1)$$

so f is continuous from right at -1 and continuous from left at 1 .

Theorem 84 If f and g are continuous at a and c is a constant, then the following functions are also continuous at a :

$$1. f \pm g$$

2. cf
3. fg
4. f/g if $g(a) \neq 0$
5. f^n , n a positive integer
6. $\sqrt[n]{f}$, provided it is defined on an open interval containing c , where n is a positive integer.
7. Any polynomial is continuous everywhere, that is it is continuous on $\mathbb{R} = (-\infty, \infty)$.
8. Any rational function is continuous wherever it is defined; that is it is continuous on its domain.

It is worth noting that all six trigonometric functions are continuous wherever they are defined.

Theorem 85 If f is continuous at a and g is continuous at $f(a)$, then the composite $g \circ f$ is continuous at a .

Theorem 86 (The Intermediate Value Theorem) Suppose that f is continuous on $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

Exercise 87 Continuous Functions

1. Find the numbers if any where the function is discontinuous.

1.1. $f(x) = 2x^3 - 3x^2 + 4$

1.3. $f(x) = \frac{x^2 - 3x + 2}{x^2 - 2x}$

1.2. $f(x) = \frac{1}{x-2}$

1.4. $f(x) = \begin{cases} 2x-1 & \text{if } x \leq 0 \\ 1 & \text{if } x > 0 \end{cases}$

2. Find the value of k that will make f continuous on $(-\infty, \infty)$

$$f(x) = \begin{cases} \frac{x^2 - 4}{x + 2} & \text{if } x \neq -2 \\ k & \text{if } x = -2 \end{cases}$$

3. Let

$$f(x) = \begin{cases} x+2 & \text{if } x \leq 1 \\ kx^2 & \text{if } x > 1 \end{cases}$$

Find the value of k that will make f continuous on $(-\infty, \infty)$.

4. Let

$$f(x) = \begin{cases} ax+b & \text{if } x < 1 \\ 4 & \text{if } x = 1 \\ 2ax-b & \text{if } x > 1 \end{cases}$$

Find the values of a and b that will make f continuous on $(-\infty, \infty)$.

5. For what value of the constant c is the function f continuous on $(-\infty, \infty)$?

$$f(x) = \begin{cases} cx^2 + 2x & \text{if } x < 2 \\ x^3 - cx & \text{if } x \geq 2 \end{cases}$$

6. Find the value of a and b that make the function f continuous everywhere.

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2} & \text{if } x < 2 \\ ax^2 - bx + 3 & \text{if } 2 \leq x < 3 \\ 2x - a + b & \text{if } x \geq 3 \end{cases}$$

7. Find the value of c that will make f continuous on $(-\infty, \infty)$.

$$f(x) = \begin{cases} \frac{\sin 2x}{x} & \text{if } x \neq 0 \\ c & \text{if } x = 0 \end{cases}$$

8. For what values of A is

$$f(x) = \begin{cases} A^2x^2 & \text{if } x \leq 2 \\ (1 - A)x & \text{if } x > 2 \end{cases}$$

continuous at 2?

9. Find the value of k that will make f continuous on $(-\infty, \infty)$

$$f(x) = \begin{cases} kx + 1 & \text{if } x \leq 2 \\ kx^2 - 3 & \text{if } x > 2 \end{cases}$$

10. Find the value of c that will make f continuous at $x = 0$

$$f(x) = \begin{cases} x \cot kx & \text{if } x < 0 \\ x^2 + c & \text{if } x \geq 0 \end{cases}$$

3 Differentiation

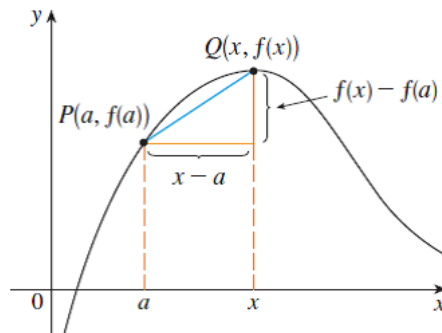
The problem of finding the tangent line to a curve and the problem of finding the velocity of an object both involve finding the same limit. This special type of limit is called the **derivative**.

3.1 Tangents

If a curve C has equation $y = f(x)$ and we want to find the tangent line to C , at a point $P(a, f(a))$, then we consider a nearby point $Q(x, f(x))$ where $x \neq a$, and compute the slope of a secant line PQ :

$$m_{pq} = \frac{f(x) - f(a)}{x - a}$$

Then we let Q approach P along the curve C by letting x approach a . If m_{pq} approaches a number m , then we define the tangent t to be the line through P with slope (gradient) m . This amounts to saying that the tangent line is the limiting position of the secant line PQ as Q approaches P .



Definition 88 The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope (gradient)

$$m_{pq} = m_T = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

provided that this limit exists.

Example 89 Find the equation of the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$.

Solution 90 Here we have $a = 1$ and $f(x) = x^2$, so the slope is

$$\begin{aligned}
 m &= \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} \\
 &= \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{x - 1} = \lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2
 \end{aligned}$$

Using the point slope form of the equation of a straight line we find that the equation of the tangent line at $(1, 1)$ is

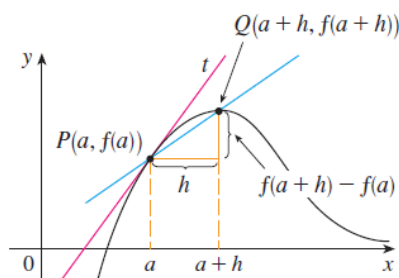
$$y - 1 = 2(x - 1) = 2x - 1.$$

Another more convenient way of expressing the slope (gradient) of a secant line is

$$m_{pq} = \frac{f(a+h) - f(a)}{h}$$

where $h = x - a$ and $x = a + h$. As such the slope (gradient) of a tangent line in the definition above becomes

$$m_T = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$



Example 91 Find the equation of the tangent line to the hyperbola $y = \frac{3}{x}$ at the point $(3, 1)$.

Solution 92 Exercise.

3.2 The Derivative as a Function

We now investigate the derivative as a function derived from f by considering the limit at each point x in the domain of f .

Definition 93 The **derivative** of a function $f(x)$ with respect to the variable x is the function f' whose value at x is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists.

By writing $z = x + h$ so that $h = z - x$, we have an alternative definition for the derivative which is sometimes more convenient to use when finding a derivative of a function.

$$f'(x) = \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

3.2.1 Calculating Derivatives from the Definition

The process of calculating a derivative is called **differentiation**.

Example 94 Use the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find the derivative of

$$f(x) = \frac{x}{x-1} \text{ and } f(x) = \frac{1-x}{2+x}$$

Solution 95 Exercise.

Example 96 Use the alternative definition $f'(x) = \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$ to find the derivative of $f(x) = \sqrt{x}$.

Solution 97 Exercise.

1. Differentiate the function by forming the difference quotient

$$\frac{f(x+h) - f(x)}{h}$$

and taking the limit as h tends to 0.

- | | |
|----------------------------|---------------------------------|
| (a) $f(x) = 2 - 3x$ | (b) $f(x) = \sqrt{x-1}$ |
| (c) $f(x) = \frac{1}{x^2}$ | (d) $f(x) = \frac{1}{\sqrt{x}}$ |
| (e) $f(x) = x + \sqrt{x}$ | |

2. Use the formula

$$f'(x) = \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

to find the derivative of the function

- | | |
|----------------------------|----------------------------|
| (a) $f(x) = \frac{1}{x+2}$ | (b) $f(x) = x^2 - 3x + 4$ |
| (c) $g(x) = 1 + \sqrt{x}$ | (d) $f(x) = \frac{x}{x-1}$ |

Notation 98 There are many ways to denote the derivative of a function $y = f(x)$ where the independent variable is x and the dependent variable is y . Some common notations for the derivative are

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = D(f)(x) = D_x f(x) = \dot{y}$$

The symbols D and d/dx are called **differentiation operators** because they indicate the operation of **differentiation**.

If we want to indicate the value of a derivative $\frac{dy}{dx}$ in Leibniz notation at a specific number a , we use the notation $\left. \frac{dy}{dx} \right|_{x=a}$ which is a synonym for $f'(a)$.

Definition 99 A function f is differentiable at a if $f'(a)$ exists. It is differentiable on an open interval (a, b) [or (a, ∞) or $(-\infty, a)$ or $(-\infty, \infty)$] if it is differentiable at every number in the interval.

Example 100 Show that the function $y = |x|$ is not differentiable at $x = 0$.

Solution 101 Exercise.

3.2.2 Differentiable Functions are continuous.

A function is continuous at every point where it has a derivative.

Theorem 102 If f is differentiable at a , then f is continuous at a .

Proof. Given that $f'(a)$ exists, we must show that

$$\lim_{x \rightarrow a} f(x) = f(a)$$

or equivalently that

$$\lim_{h \rightarrow 0} f(a+h) = f(a)$$

If $h \neq 0$, then

$$\begin{aligned} f(a+h) &= f(a) + (f(a+h) - f(a)) \\ &= f(a) + \frac{f(a+h) - f(a)}{h} \cdot h. \end{aligned}$$

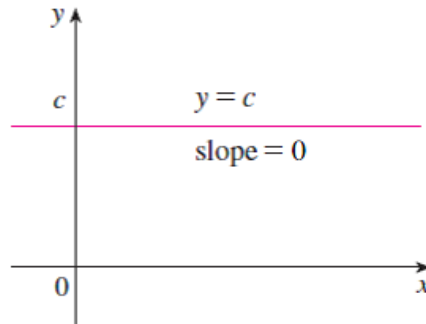
Now take the limits as $h \rightarrow 0$

$$\begin{aligned} \lim_{h \rightarrow 0} f(a+h) &= \lim_{h \rightarrow 0} f(a) + \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \cdot \lim_{h \rightarrow 0} h \\ &= f(a) + f'(a) \cdot 0 \\ &= f(a). \end{aligned}$$

■

HOW CAN A FUNCTION FAIL TO BE DIFFERENTIABLE?

In general, if the graph of a function f has a "corner" or a "kink" in it, then the graph of f has no tangent at this point and f is not differentiable there.



3.3 Differentiation Rules

3.3.1 Powers, Multiples, Sums, and Differences

A simple rule of differentiation is that the derivative of every constant function is zero.

Derivative of a Constant Function

If f has the constant value $f(x) = c$, then $\frac{df}{dx} = \frac{d}{dx}(c) = 0$

Example 103 $\frac{d}{dx}(3) = 0$

Power Rule (General Version)

If n is any real number, then $\frac{d}{dx}x^n = nx^{n-1}$ for all x where x^n and x^{n-1} are defined.

Example 104 .

(a) $\frac{d}{dx}(x^4) = 4x^3$

(b) If $y = x^{100}$, then $\frac{dy}{dx} = 100x^{99}$

(c) If $y = \sqrt[3]{x^2}$, find $\frac{dy}{dx}$.

$$y = x^{2/3} \text{ and so, } \frac{dy}{dx} = \frac{2}{3}x^{2/3-1} = \frac{2}{3}x^{-1/3}$$

(d) Find the derivative of $y = \sqrt{x^{2+\pi}}$. The solution is left as an **exercise**.

Example 105 Find the equations of the tangent line and normal line to the curve $y = x\sqrt{x}$ at the point $(1, 1)$.

Solution 106 The derivative of $y = f(x) = x\sqrt{x} = x \cdot x^{1/2} = x^{3/2}$ is

$$\frac{dy}{dx} = f'(x) = \frac{3}{2}x^{1/2} = \frac{3}{2}\sqrt{x}$$

So the slope of the tangent line at $(1, 1)$ is $f'(1) = \frac{3}{2}$. Therefore the equation of the tangent line is

$$y - 1 = \frac{3}{2}(x - 1) \text{ or } y = \frac{3}{2}x - \frac{1}{2}$$

The normal line is perpendicular to the tangent line, so its slope is the negative reciprocal of $\frac{3}{2}$, that is $-\frac{2}{3}$. Thus the equation of the normal line is

$$y - 1 = -\frac{2}{3}(x - 1) \text{ or } y = -\frac{2}{3}x + \frac{5}{3}$$

The Constant Multiple Rule

If f is a differentiable function and c is a constant, then $\frac{d}{dx}[cf(x)] = c \cdot f'(x)$

The sum/ Difference Rule

If f and g are both differentiable functions of x , then

$$\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}f(x) \pm \frac{d}{dx}g(x)$$

Example 107

$$\begin{aligned} \frac{d}{dx}(x^8 + 12x^5 - 4x^4 + 10x^3 - 6x + 5) &= \frac{d}{dx}(x^8) + 12\frac{d}{dx}(x^5) - 4\frac{d}{dx}(x^4) \\ &\quad + 10\frac{d}{dx}(x^3) - 6\frac{d}{dx}(x) + \frac{d}{dx}(5) \\ &= 8x^7 + 12(5x^4) - 4(4x^3) + 10(3x^2) - 6 + 0 \\ &= 8x^7 + 60x^4 - 16x^3 + 30x^2 - 6 \end{aligned}$$

Does

$$y = x^4 - 2x^2 + 2$$

have any horizontal tangents? If so where?

Solution 108 The horizontal tangents, if any, occur where the gradient $\frac{dy}{dx} = 0$.

We have

$$\frac{dy}{dx} = 4x^3 - 4x$$

Now solve the equation $\frac{dy}{dx} = 0$ for x :

$$\begin{aligned} 4x^3 - 4x &= 0 \\ 4x(x^2 - 1) &= 0 \\ x &= 0, -1, 1. \end{aligned}$$

The curve

$$y = x^4 - 2x^2 + 2$$

has horizontal tangents at $x = -1$, $x = 0$ and $x = 1$.

Theorem 109 (The Product Rule) If f and g are differentiable functions, then

$$\frac{d}{dx}[f(x) \cdot g(x)] = f(x) \frac{d}{dx}[g(x)] + g(x) \frac{d}{dx}[f(x)]$$

Proof. We form the difference quotient

$$\begin{aligned} & \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ = & \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h} \end{aligned}$$

and write it as

$$f(x+h) \left[\frac{g(x+h) - g(x)}{h} \right] + g(x) \left[\frac{f(x+h) - f(x)}{h} \right]$$

Since f is differentiable at x , we know that f is continuous at x and thus

$$\lim_{h \rightarrow 0} f(x+h) = f(x)$$

Since

$$\lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = g'(x)$$

and

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$$

we obtain

$$\begin{aligned} & \lim_{h \rightarrow 0} f(x+h) \lim_{h \rightarrow 0} \left[\frac{g(x+h) - g(x)}{h} \right] + g(x) \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} \right] \\ = & f(x) \frac{d}{dx} [g(x)] + g(x) \frac{d}{dx} [f(x)] \end{aligned}$$

■

Theorem 110 (The Quotient Rule) *If f and g are differentiable functions and $g(x) \neq 0$, then*

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

Proof.

$$\begin{aligned} \frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] &= \lim_{h \rightarrow 0} \left[\frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} \right] \\ &= \lim_{h \rightarrow 0} \left[\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)} \right] \cdot \frac{1}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{g(x)f(x+h) - f(x)g(x+h)}{g(x)g(x+h)} \right] \cdot \frac{1}{h} \end{aligned}$$

Subtract and add $g(x)f(x)$ in the numerator to get,

$$\begin{aligned}\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] &= \lim_{h \rightarrow 0} \left[\frac{g(x)f(x+h) - g(x)f(x) + g(x)f(x) - f(x)g(x+h)}{g(x)g(x+h)} \right] \cdot \frac{1}{h} \\ &= \lim_{h \rightarrow 0} \left\{ \frac{g(x) \left[\frac{f(x+h) - f(x)}{h} \right] - f(x) \left[\frac{g(x+h) - g(x)}{h} \right]}{g(x)g(x+h)} \right\}\end{aligned}$$

Taking the limits in the numerator and the denominator now gives the **Quotient Rule**. ■

Example 111 Find the derivative of

$$y = (x^2 + 1)(x^3 + 3)$$

Solution 112 We use the product rule : $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$.

$$\begin{aligned}\frac{dy}{dx} &= (x^2 + 1)3x^2 + (x^3 + 3x)(2x) \\ &= 3x^4 + 3x^2 + 2x^4 + 6x \\ &= 5x^4 + 3x^2 + 6x\end{aligned}$$

Example 113 Find the derivative of

$$y = \frac{t^2 - 1}{t^3 + 1}$$

Solution 114 We use quotient rule: $\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \cdot u' - u \cdot v'}{v^2}$.

$$\begin{aligned}\frac{dy}{dt} &= \frac{(t^3 + 1)(2t) - (t^2 - 1)(3t^2)}{(t^3 + 1)^2} \\ &= \frac{2t^4 + 2t - 3t^4 + 3t^2}{(t^3 + 1)^2} \\ &= \frac{-t^4 + 3t^2 + 2t}{(t^3 + 1)^2}\end{aligned}$$

We summarise the rules discussed above as follows:

$$\begin{array}{ll}\frac{d}{dx}(c) = 0 & \frac{d}{dx}(x^n) = nx^{n-1} \\ (f \cdot g)' = fg' + gf' & (cf)' = cf' \\ (f \pm g)' = f' \pm g' & \left(\frac{f}{g} \right)' = \frac{gf' - fg'}{g^2}\end{array}$$

Higher Order Derivatives

If $y = f(x)$ is a differentiable function, then its derivative $f'(x)$ is also a function. If f' is also differentiable, we get a new function of x denoted by f'' which is called the second derivative of f . It is written in several ways:

$$f''(x) = \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{dy'}{dx} = y'' = D_x^2 f(x) = \dots$$

Higher order derivatives may also be found

$$f'(x), f''(x), f'''(x), f^{(4)}(x), \dots, f^{(n)}(x)$$

Exercise 115 .

1. Differentiate the function.

$$1.1. f(x) = x^3 - 4x + 6$$

$$1.2. f(x) = (x-2)(2x+3)$$

$$1.3. S(p) = \sqrt{p} - p$$

$$1.4. y = x^{5/3} - x^{2/3}$$

$$1.5. f(x) = x^2(1-2x)$$

$$1.6. y = \frac{x^3}{1-x^2}$$

$$1.7. f(x) = \frac{x}{x + \frac{c}{x}}$$

$$1.8. f(x) = -4x^{10}$$

$$1.9. v = \left(\sqrt{x} + \frac{1}{\sqrt[3]{x}} \right)^2$$

$$1.10. y = \frac{x^3 + 3x}{x^2 - 1}$$

$$1.11. g(x) = (9x^8 - 8x^9) \left(x + \frac{1}{x} \right)$$

$$1.12. y = \frac{x^2 + 4x + 3}{\sqrt{x}}$$

$$1.13. y = \frac{v^3 - 2v\sqrt{v}}{v}$$

$$1.14. f(x) = \frac{ax - b}{cx - d}$$

2. Find $f'(0)$ and $f'(1)$.

$$2.1. f(x) = \frac{1}{x-2}$$

$$2.3. f(x) = \frac{2x^2 + x + 1}{x^2 + 2x + 1}$$

$$2.2. f(x) = x^2(x+1)$$

$$2.4. f(x) = \frac{ax+b}{cx+d}$$

3. Find $f'(0)$ given that $h(0) = 3$ and $h'(0) = 2$

$$3.1. f(x) = xh(x)$$

$$3.2. f(x) = h(x) + \frac{x}{h(x)}$$

4. Find the equation of the tangent line at the point $(c, f(c))$

$$f(x) = \frac{x}{x+2}; c = -4$$

5. Find the equation of the tangent line to the graph of

$$f(x) = \frac{(2x^2 + 1)(x^3 - 1)}{x^2 + 4}$$

at the point where $x = 1$.

6. Find the indicated derivatives

6.1. $\frac{d}{dx}(3x - e)$

6.2. $\frac{d}{dt}\left(\frac{2t^3 + 1}{t^4}\right)$

7. Find the second derivative

7.1. $f(x) = 7x^4 - 6x^5$

7.3. $y = x^2\left(x + \frac{3}{x}\right)$

7.2. $y = \frac{x^2 - 3\pi}{x}$

7.4. $f(y) = y^{-1} + 3y^{-3}$

8. Find $\frac{d^3y}{dx^3}$.

8.1. $y = \frac{1}{3}x^3 + \frac{1}{2}x^2 + x + 1$

8.3. $y = (\sqrt{5} + 5x)^2$

8.2. $y = x^3 - \frac{1}{x^3}$

8.4. $y = \frac{x^4 - 2}{x}$

3.3.2 Derivatives of Trigonometric Functions

From our discussion on trigonometric functions, we recall that

$$\begin{aligned}\sin(x+h) &= \sin x \cos h + \cos x \sin h \\ \cos(x+h) &= \cos x \cos h - \sin x \sin h \\ \lim_{x \rightarrow 0} \frac{\sin x}{x} &= 1 \\ \lim_{x \rightarrow 0} \frac{\cos x - 1}{x} &= 0\end{aligned}$$

Theorem 116 (Derivative of $\sin x$)

$$\frac{d}{dx}(\sin x) = \cos x$$

Proof. . Let $f(x) = \sin x$. Then

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sin x (\cos h - 1) + \cos x \sin h}{h} \\
 &= \lim_{h \rightarrow 0} \left(\sin x \cdot \frac{\cos h - 1}{h} \right) + \lim_{h \rightarrow 0} \left(\cos x \cdot \frac{\sin h}{h} \right) \\
 &= \sin x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \cos x \lim_{h \rightarrow 0} \frac{\sin h}{h} \\
 &= \sin x \cdot 0 + \cos x \cdot 1 \\
 &= \cos x.
 \end{aligned}$$

■

Exercise 117 Show that $\frac{d}{dx}(\cos x) = -\sin x$.

Derivatives of the six trigonometric ratios are given below:

$$\begin{array}{ll}
 \frac{d}{dx}(\sin x) = \cos x & \frac{d}{dx}(\csc x) = -\csc x \cot x \\
 \frac{d}{dx}(\cos x) = -\sin x & \frac{d}{dx}(\sec x) = \sec x \tan x \\
 \frac{d}{dx}(\tan x) = \sec^2 x & \frac{d}{dx}(\cot x) = -\csc^2 x
 \end{array}$$

Example 118 Differentiate $y = \frac{\sec x}{1 + \tan x}$

$$\begin{aligned}
 \frac{dy}{dx} &= \frac{(1 + \tan x) \frac{d}{dx}(\sec x) - \sec x \frac{d}{dx}(1 + \tan x)}{(1 + \tan x)^2} \\
 &= \frac{(1 + \tan x) \sec x \tan x - \sec x \sec^2 x}{(1 + \tan x)^2} \\
 &= \frac{\sec x [\tan x + \tan^2 x - \sec^2 x]}{(1 + \tan x)^2} \\
 &= \frac{\sec x (\tan x - 1)}{(1 + \tan x)^2}.
 \end{aligned}$$

Exercise 119

1. Differentiate

$$(a) \quad f(x) = 3x^2 - 2\cos x$$

$$(b) \quad y = \sqrt{x} \sin x$$

$$(c) \quad y = \sin x + \frac{1}{2} \cot x$$

$$(d) \quad y = 2 \sec x - \csc x$$

$$(e) \quad h(\theta) = \theta \csc \theta - \cot \theta$$

$$(f) \quad y = \frac{\cos x}{1 - \sin x}$$

$$(g) \quad y = \frac{1 - \sec x}{\tan x}$$

$$(h) \quad y = x^2 \sin x \tan x$$

2. Prove using the definition of the derivative, that if $f(x) = \cos x$, then $f'(x) = -\sin x$.

3.4 The Chain Rule

Suppose we wish to differentiate

$$F(x) = (x^2 + 1)^{120}$$

We may view F as a composition of

$$g(x) = x^{120} \text{ and } f(x) = x^2 + 1.$$

$$F(x) = (g \circ f)(x) = g[f(x)] = [f(x)]^{120} = (x^2 + 1)^{120}$$

Theorem 120 (The Chain Rule) *If $f(u)$ is differentiable at the point $u = g(x)$ and $g(x)$ is differentiable at x , then the composite function*

$$(f \circ g)(x) = f(g(x))$$

is differentiable at x and

$$(f \circ g)'(x) = f'[g(x)] \cdot g'(x)$$

In Leibniz's notation, if

$$y = f(x)$$

and

$$u = g(x)$$

then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

where $\frac{dy}{du}$ is evaluated at $u = g(x)$.

NOTE: Label the composite function

$$h(x) = \underbrace{f}_{\text{"Outside function"}} \left(\underbrace{g(x)}_{\text{"Inside function"}} \right)$$

then $h'(x)$ is just the derivative of the "outside function" evaluated at the "inside function" times the derivative of the "inside function".

Example 121 *Differentiate*

$$y = (x^2 + 1)^{120}$$

Solution 122

$$\frac{dy}{dx} = 120(x^2 + 1)^{119} \cdot \frac{d}{dx}(x^2 + 1) = 240x(x^2 + 1)^{119}$$

NOTE: If we let $u = x^2 + 1$ and $y = u^{120}$, then $\frac{du}{dx} = 2x$ and $\frac{dy}{du} = 120u^{119}$. So

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = 2x \left[120(x^2 + 1)^{119} \right] = 240x(x^2 + 1)^{119}.$$

Example 123 *Find the derivative*

$$(a) \ F(x) = \sqrt{x^2 + 1}$$

$$(b) \ y = \sin 2x^3$$

$$(c) \ y = \sin^3 2x$$

Solution 124 *Exercise.*

As observed in the previous example, we can in general combine the Power Rule with the Chain Rule to get:

If n is any number and $u = g(x)$ is differentiable, then

$$\frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}$$

Alternately,

$$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1} \cdot g'(x)$$

Example 125 *Find $f'(x)$ if $f(x) = \left(\frac{x-2}{2x+1}\right)^9$*

Example 126 *Find $\frac{dy}{dx}$ if $y = \frac{1}{\sqrt[3]{x^2 + x + 1}}$*

Example 127 *Differentiate $y = \sin(\cos(\tan x))$*

Exercise 128

Find the Derivative of the Function

1. $y = (2x + 4)^3$
2. $f(x) = \sqrt{x + \cos x}$
3. $f(t) = (2t^3 - t)^{-3}$
4. $g(x) = \frac{1}{(x^3 - 2x^2 + 1)^5}$
5. $y = \left(t + \frac{2}{t}\right)^6$
6. $h(x) = (2x - 1)^2 (x^2 + 1)^3$
7. $g(t) = \sqrt{t - 2} + \sqrt{4 - t}$
8. $f(x) = \left(\frac{x^2 + 3}{x}\right)^{-2}$
9. $y = \frac{x}{\sqrt{x^2 - x - 2}}$
10. $y(s) = \left(1 + \sqrt{1 + s^2}\right)^5$
11. $g(x) = \left(\frac{2x^2 - 1}{2x + 5}\right)^{1/3}$
12. $f(x) = \sin^3 x$
13. $y = \cot^2(\sin \theta)$
14. $g(x) = \tan^2(x^2 + x)$
15. $f(x) = \sqrt{\sin 2x - \cos 2x}$
16. $f(x) = \sqrt{2 + 3 \tan 2x}$
17. $y = x \sin \frac{1}{x}$
18. $y = \sqrt{\sin(\cos 2x)}$
19. $y = \sqrt{x + \sqrt{x + \sqrt{x}}}$
20. $f(s) = \sqrt{\frac{s^2 + 1}{s^2 + 4}}$
21. $y = \left(\frac{1 - \cos 2x}{1 + \cos 2x}\right)^4$
22. $y = \cos(x^2 - 3x + 1) + \tan\left(\frac{2}{x}\right)$

3.5 Implicit Differentiation

1. Differentiate both sides of the equation with respect to x , treating y as a differentiable function of x .
2. Collect the terms with $\frac{dy}{dx}$ (or y') on one side of the equation and solve for $\frac{dy}{dx}$ (or y').

Example 129 Find $\frac{dy}{dx}$ if $y^2 = x^2 + \sin xy$.

Solution 130

$$\begin{aligned}
 \frac{d}{dx}(y^2) &= \frac{d}{dx}(x^2) + \frac{d}{dx}(\sin xy) \\
 2y \frac{dy}{dx} &= 2x + (\cos xy) \frac{d}{dx}(\sin xy) \\
 2y \frac{dy}{dx} &= 2x + (\cos xy) \left(y + x \frac{dy}{dx}\right) \\
 2y \frac{dy}{dx} &= 2x + y(\cos xy) + x(\cos xy) \frac{dy}{dx} \\
 2y \frac{dy}{dx} - x(\cos xy) \frac{dy}{dx} &= 2x + y(\cos xy) \\
 [2y - x(\cos xy)] \frac{dy}{dx} &= 2x + y(\cos xy) \\
 \frac{dy}{dx} &= \frac{2x + y \cos xy}{2y - x \cos xy}
 \end{aligned}$$

1. Find $\frac{dy}{dx}$ by implicit differentiation

- | | |
|--|---------------------------------------|
| (a) $xy^2 + yx^2 - 2 = 0$ | (b) $x^2y + 2xy^2 - x + 3 = 0$ |
| (c) $\sqrt{x} + \sqrt{y} = 1$ | (d) $\tan^2(x^3 + y^3) = xy$ |
| (e) $\sqrt{1 + \cos^2 y} = xy$ | (f) $1 + x = \sin(xy^2)$ |
| (g) $y^5 + x^2y^3 = 1 + x^4y$ | (h) $y \sin(x^2) = x \sin(y^2)$ |
| (i) $\tan\left(\frac{x}{y}\right) = x + y$ | (j) $\tan(x - y) = \frac{y}{1 + x^2}$ |

3.5.1 Higher Order Derivatives

Implicit differentiation can be used to find higher derivatives.

Example 131 Find $\frac{d^2y}{dx^2}$ if $2x^3 - 3y^2 = 8$

Solution 132

$$\begin{aligned}\frac{d}{dx}(2x^3 - 3y^2) &= \frac{d}{dx}(8) \\ 6x^2 - 6yy' &= 0 \\ y' &= \frac{x^2}{y}, \quad \text{when } y \neq 0\end{aligned}$$

We apply the Quotient Rule to find y'' .

$$y'' = \frac{d}{dx}\left(\frac{x^2}{y}\right) = \frac{2xy - x^2y'}{y^2} = \frac{2x}{y} - \frac{x^2}{y^2} \cdot y'$$

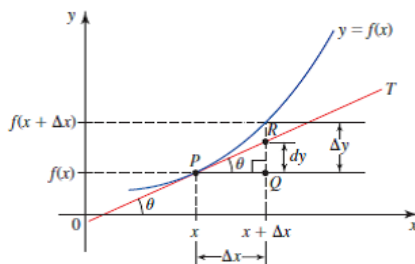
Finally, we substitute $y' = \frac{x^2}{y}$ to express y'' in terms of x and y .

$$y'' = \frac{2x}{y} - \frac{x^2}{y^2} \left(\frac{x^2}{y}\right) = \frac{2x}{y} - \frac{x^4}{y^3} \quad \text{when } y \neq 0$$

1. Show that $\frac{d^2y}{dx^2} = -48\frac{x^2}{y^7}$ if $x^4 + y^4 = 16$

2. Find $\frac{d^2y}{dx^2}$

- $xy + x^3 = 4$
- $x^3 - y^3 = 8$
- $\sin x + \cos y = 1$
- $\tan y - xy = 0$



3.6 Differentials

To find a quick way of estimating the change in y , Δy , due to small change in x , Δx , let us look at the graph below.

The tangent line T lies close to the graph of f near the point of tangency at P . Therefore, if Δx is small, the y -coordinate of the point R on T is a good approximation of $f(x + \Delta x)$. Equivalently, the quantity dy is a good approximation of Δy . Now consider the right triangle ΔPQR . We have

$$\frac{dy}{\Delta x} = \tan \theta$$

or $dy = (\tan \theta) \Delta x$. But the derivative of f gives the slope of the tangent line T , so we have $\tan \theta = f'(x)$. Therefore,

$$dy = f'(x) \Delta x$$

The quantity dy is called the differential of y .

Definition 133 (Differential) Let $y = f(x)$ where f is a differentiable function. Then

1. The **differential** dx of the independent variable x is $dx = \Delta x$, where Δx is an increment in x .
2. The **differential** dy of the dependent variable y is

$$dy = f'(x) \Delta x = f'(x) dx.$$

Example 134 Find the differential of the $f(x) = 2 \sin x + 3 \cos x$ at $x = \frac{\pi}{4}$

Solution 135 $dy = f'(x) \Delta x = f'(x) dx$

$$f'(x) = 2 \cos x - 3 \sin x$$

$$dy = (2 \cos x - 3 \sin x) dx$$

$$\text{at } x = \frac{\pi}{4}, dy = 2 \cos\left(\frac{\pi}{4}\right) - 3 \sin\left(\frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2} dx$$

3.7 Transcendental Functions

Functions which are not algebraic like trigonometric functions, exponential functions, logarithmic functions, hyperbolic functions etc. are examples of transcendental functions.

3.7.1 Exponential and Logarithmic functions

Definition 136 An (common) exponential function is a function of the form

$$f(x) = a^x$$

where a is a positive constant.

Definition 137 A (common) logarithmic function is a function of the form

$$f(x) = \log_a x$$

where the base a is a positive constant. It is an inverse of the exponential function.

The number e is an irrational number and has the approximation $e = 2.718181828 \dots$. Later, it will be shown that

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e,$$

which is equivalent to

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = e.$$

The Natural Exponential Function may be denoted by

$$\exp(x) = e^x.$$

Theorem 138 *Exponential Laws and Properties*

Let x and y be real numbers. Then

$$(a) \quad e^x \cdot e^y = e^{x+y}$$

$$(b) \quad \frac{e^x}{e^y} = e^{x-y}$$

$$(c) \quad (e^x)^y = e^{xy}$$

$$(d) \quad \ln e^x = x \text{ for } x \in (-\infty, \infty)$$

$$(e) \quad e^{\ln x} = x, \text{ for } x \in (0, \infty)$$

Theorem 139 (The Derivative of Exponential Function)

$$\frac{d}{dx}(e^x) = e^x.$$

Proof. We know that the derivative of a function $f(x)$ with respect to the variable x is the function

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \text{provided the limit exists.}$$

Now,

$$\begin{aligned}
 \frac{d}{dx}[e^x] &= \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{e^x e^h - e^x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{e^x(e^h - 1)}{h} \\
 &= e^x \lim_{h \rightarrow 0} \frac{(e^h - 1)}{h} \\
 &\quad \boxed{\text{SUBSTITUTE:}} \\
 &\quad \boxed{u = e^h - 1 \implies e^h = u + 1 \implies h = \ln(u + 1)} \\
 &= e^x \lim_{u \rightarrow 0} \frac{u}{\ln(u + 1)} \\
 &= e^x \lim_{u \rightarrow 0} \frac{1/u \cdot (u)}{1/u \ln(u + 1)} \\
 &= e^x \lim_{u \rightarrow 0} \frac{1}{\ln(u + 1)^{1/u}} \\
 &= e^x \cdot \frac{1}{\ln \left[\lim_{u \rightarrow 0} (1 + u)^{1/u} \right]} \\
 &= e^x \cdot \frac{1}{\ln e} \\
 &= e^x.
 \end{aligned}$$

■

In General, we have $\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$.

Example 140 Find the derivative

$$(a) \quad y = e^{\sqrt{x+1}}$$

Solution 141

$$\begin{aligned}
 \frac{dy}{dx} &= e^{\sqrt{x+1}} \frac{d}{dx}(x+1)^{1/2} \\
 &= e^{\sqrt{x+1}} \left(\frac{1}{2} \right) (x+1)^{-1/2} (1) \\
 &= \frac{e^{\sqrt{x+1}}}{2\sqrt{x+1}}
 \end{aligned}$$

3.7.2 Natural Logarithmic function

$$\log_e x = \ln x$$

Laws of Logarithms

Theorem 142 Let A and B be positive numbers and let n be a rational number. Then

$$(a) \ln 1 = 0$$

$$(b) \ln AB = \ln A + \ln B$$

$$(c) \ln \frac{A}{B} = \ln A - \ln B$$

$$(d) \ln A^n = n \ln A$$

Definition 143 The natural exponential function denoted by $\exp x$ is the function that satisfy the following conditions:

$$1. \ln(\exp x) = x, \forall x \in (-\infty, \infty) \text{ i.e. } \ln e^x = x$$

$$2. \exp(\ln x) = x, \forall x \in (0, \infty) \text{ i.e. } e^{\ln x} = x$$

Definition 144 If x is any real number, then

$$e^x = y \iff \ln y = x$$

Example 145 Expand the expression.

$$(a) \ln \frac{x^2 + 1}{\sqrt{x}}$$

$$(b) \ln \frac{x^3 \cos^2 \pi x}{\sqrt{x^2 + 3}}$$

Solution 146 Exercise.

Theorem 147 (Derivative of the Natural Logarithmic function) Let $f(x) = \ln x$, then

$$f'(x) = \frac{d}{dx}(\ln x) = \frac{1}{x}.$$

Proof.

$$\begin{aligned} \frac{d}{dx}[\ln x] &= \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x}{h} \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \ln \left(\frac{x+h}{x} \right) \\ &= \lim_{h \rightarrow 0} \frac{1}{h} \ln \left(1 + \frac{h}{x} \right) \\ &\quad \boxed{\begin{array}{l} \text{SUBSTITUTE:} \\ u = \frac{h}{x} \implies h = ux \end{array}} \\ &= \lim_{u \rightarrow 0} \frac{1}{ux} \ln(1+u) \\ &= \frac{1}{x} \lim_{u \rightarrow 0} \left[\ln(1+u)^{1/u} \right] \\ &= \frac{1}{x} \ln \left[\lim_{u \rightarrow 0} (1+u)^{1/u} \right] \\ &= \frac{1}{x} \ln e \\ &= \frac{1}{x}. \end{aligned}$$

■

In general, we observe that if u is a differentiable function of x , Chain rule gives us,

$$\frac{d}{dx} \ln u = \frac{1}{u} \frac{du}{dx}, u > 0,$$

or equivalently,

$$\frac{d}{dx} \ln[f(x)] = \frac{f'(x)}{f(x)}$$

Example 148 Find the derivative of

(i) $f(x) = \ln(2x^2 + 1)$

(ii) $g(x) = x^2 \ln 2x$

(iii) $f(x) = \ln \sqrt{x^2 + 1}$

(iv) $y = \ln \left[\frac{x^2 (2x^2 + 1)^3}{\sqrt{5 - x^2}} \right]$

(v) $y = \sec[\ln(2x + 3)]$

$$(vi) \ y = \ln \left(\frac{\sqrt{\sin \theta \cos \theta}}{1 + 2 \ln \theta} \right)$$

Solution 149 *Exercise*

Logarithmic Differentiation

This method is especially useful for differentiating functions involving, products, quotients and/or powers that can be simplified using logarithms.

Example 150 *Find the derivative of*

$$y = x^{\cos x}$$

Solution 151 *We introduce the natural logarithm on both sides of the equation:*

$$\begin{aligned} \ln y &= \ln x^{\cos x} \\ \ln y &= \cos x \ln x \\ \frac{1}{y} \frac{dy}{dx} &= -\sin x \ln x + \frac{\cos x}{x} \\ \frac{dy}{dx} &= x^{\cos x} \left[\frac{\cos x}{x} - \sin x \ln x \right] \end{aligned}$$

Example 152 *Differentiate*

$$(a) \ y = \frac{(2x-1)^3}{\sqrt{3x+1}}$$

$$(b) \ y = \sqrt[3]{\frac{x(x+1)(x-2)}{(x^2+1)(2x+3)}}$$

Solution 153 *Exercise.*

Exercise

Differentiate the function.

1. $f(x) = \ln(1-x^2)$
2. $g(x) = \ln(x^2+4)^2$
7. $f(x) = \ln(x+\sqrt{x^2-1})$
3. $f(x) = x(\ln x)^2$
4. $y = \frac{\ln x}{x+1}$
8. $f(t) = t \sin(\ln t)$
5. $u = \ln \sqrt{\frac{x \cos x}{(2x+1)^3}}$
6. $f(x) = \ln(\ln x)$
9. $g(t) = \ln \left| \frac{\sin t + 1}{\cos t + 2} \right|$

Use logarithmic differentiation to find the derivative of the function.

10. $y = (1-x^2)^2 (3x^2-4)^3$
11. $y = \frac{x^2 \sqrt{2x-4}}{(x+1)^2}$
12. $f(x) = \sqrt[3]{\frac{x-1}{x^2+1}}$
13. Find y'' if $y = x^x$
14. $y = \frac{\sin^2 x}{x^2 \sqrt{1+\tan x}}$
15. Find y' if $y = (x^2+2)^{\sin x}$

Use implicit differentiation to find $\frac{dy}{dx}$.

$$16. \ln y - x \ln x = -1 \quad 17. \ln(x+y) - \cos y - x^2 = 0$$

$$18. y = \ln(x^2 + y^2) \quad 19. x^y = y^x$$

$$20. \ln \frac{x}{y} + x - y^2 = 0$$

3.7.3 The Derivatives of a^x and a^u

Definition 154 Let a be a positive real number with $a \neq 1$. The exponential function with base a is the function f defined by

$$f(x) = a^x = e^{x \ln a}$$

for every real number x .

The usual exponential laws apply.

Theorem 155 (Derivatives of a^x and a^u) Let a be a positive number with $a \neq 1$, and let u be a differentiable function of x . Then

$$(a) \frac{d}{dx}(a^x) = a^x \ln a$$

$$(b) \frac{d}{dx}(a^u) = a^u \frac{du}{dx} \ln a$$

Example 156 Find the derivative

$$(a) f(x) = 3^{\sqrt{x}}$$

$$(b) y = 10^{\cos 2x}$$

Logarithmic Function With Base a

Definition 157 Logarithmic function with base a denoted by $\log_a$, is a function satisfying the relationship

$$y = \log_a x \iff x = a^y \quad (3.1)$$

Change of base formula :

$$\log_a x = \frac{\ln x}{\ln a} \quad a > 0 \text{ and } a \neq 1$$

Theorem 158 Let u be a differentiable function of x . Then

$$(a) \frac{d}{dx} \log_a |x| = \frac{1}{x \ln a}, \quad x \neq 0$$

$$(b) \frac{d}{dx} \log_a |u| = \frac{1}{u \ln a} \frac{du}{dx}, \quad u \neq 0$$

Example 159 Find the derivative

(a) $y = \log_2(3x^2 + 1)$

(b) $y = \log_3 |\tan x|$

(c) $f(x) = x^2 \log(e^{2x} + 1)$

Solution 160 *Exercise.***Exercise 161**

Differentiate:

1. $y = e^{-2x}$

2. $y = e^{2x+1}$

3. $y = e^{4 \ln x}$

4. $y = e^{\sqrt{x}} \ln \sqrt{x}$

5. $y = (x^2 - 2x + 2)e^x$

6. $f(x) = \ln(\cos e^{2x})$

7. $y = \cos\left(\frac{1 - e^{ex}}{1 + e^{2x}}\right)$

8. $f(t) = \sin(e^t) + e^{\sin t}$

9. $y = (e^{2x} - \ln 3x)^{3/2}$

10. $y = x^2 e^x - x e^{x^2}$

11. $y = (3 - 2e^{-x})^3$

12. $y = \frac{e^{2x} - 1}{e^{2x} + 1}$

13. $h(t) = t^3 - 3t$

14. $f(x) = e^{\cos x^2} \tan(e^{2x} + x)$

15. $y = 3^{-2x}$

16. $y = 4^{2x+1}$

17. $y = (\sqrt{\cos x})^x$

18. $y = 10^{\tan \theta}$

19. $y = \log_2(e^{-x} \cos \pi x)$

20. $y = 2^{3x^2}$

21. $y = x^{x^2}$

22. $g(x) = \frac{2^x}{\sqrt{3^x + 1}}$

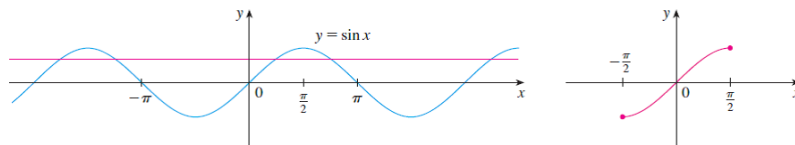
23. $y = (\sin x)^{\tan x}$

3.7.4 Inverse Trigonometric Functions

Generally speaking, the trigonometric functions, being periodic are not one-to-one and therefore do not have inverse functions. For example the graph of $y = \sin x$ is not one-to-one as it fails the horizontal line test. By restricting the domain of the function $f(x) = \sin x$ to the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ we observe that $f(x)$ becomes one-to-one and has an inverse function with domain $[-1, 1]$ and range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. This function is called the

inverse sine function or the *arcsine* function. Thus

$$y = \sin^{-1} x \iff \sin y = x \quad \text{where } -1 \leq x \leq 1 \text{ and } -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$



Definition 162 (Inverse Trigonometric Functions)

Domain

$$\begin{aligned} y = \sin^{-1} x &\iff x = \sin y & [-1, 1] \\ y = \cos^{-1} x &\iff x = \cos y & [-1, 1] \\ y = \tan^{-1} x &\iff x = \tan y & (-\infty, \infty) \\ y = \cot^{-1} x &\iff x = \cot y & (-\infty, \infty) \\ y = \sec^{-1} x &\iff x = \sec y & (-1, 1] \cup [1, \infty) \\ y = \csc^{-1} x &\iff x = \csc y & (-1, 1] \cup [1, \infty) \end{aligned}$$

Theorem 163 (Derivatives of Inverse Trigonometric Functions) *Let u be a differentiable function of x . Then*

(a)	$\frac{d}{dx} (\sin^{-1} u) = \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$
(b)	$\frac{d}{dx} (\cos^{-1} u) = \frac{-1}{\sqrt{1-u^2}} \frac{du}{dx}$
(c)	$\frac{d}{dx} (\tan^{-1} u) = \frac{1}{1+u^2} \frac{du}{dx}$
(d)	$\frac{d}{dx} (\cot^{-1} u) = \frac{-1}{1+u^2} \frac{du}{dx}$
(e)	$\frac{d}{dx} (\sec^{-1} u) = \frac{1}{ u \sqrt{u^2-1}} \frac{du}{dx}$
(f)	$\frac{d}{dx} (\csc^{-1} u) = \frac{-1}{ u \sqrt{u^2-1}} \frac{du}{dx}$

Proof. Exercise. ■

Example 164 Find the derivative of

(a) $f(x) = \cos^{-1} 3x$

(b) $y = \sec^{-1} e^{-2x}$

(c) $g(x) = \tan^{-1} \sqrt{2x+3}$

Solution 165 Exercise.

Exercise 166 Find the derivatives of the function. Simplify where possible.

1. $f(x) = \arctan \sqrt{x}$
2. $g(x) = e^x \sin^{-1}(x)$
3. $f(x) = \sqrt{\arctan 2x}$
4. $y = \sec^{-1}(2 + \cos x)$
5. $y = \sin[\sec^{-1}(\ln x)]$
6. $y = \frac{x}{\sqrt{c^2 - x^2}} - \arcsin\left(\frac{x}{c}\right)$
7. $y = \tan^{-1} \sqrt{x^2 - 1} + \csc^{-1} x$
8. $y = \tan^{-1}(\ln |x|)$
9. $y = x \sin^{-1} x + \sqrt{1 - x^2}$
10. $f(x) = \arctan \sqrt{\frac{1-x}{1+x}}$
11. $y = \tan^{-1}\left(\frac{x}{a}\right) + \ln \sqrt{\frac{x-a}{x+a}}$
12. $y = \tan^{-1}\left(x - \sqrt{1+x^2}\right)$
13. $y = \arcsin \sqrt{x}$
14. $y = \cos^{-1}(e^{2x})$

3.7.5 Hyperbolic Functions

Certain even and odd combinations of exponential functions e^x and e^{-x} arise so frequently in Mathematics and its applications that they deserve to be given specific names. These functions have the same relationship to the hyperbola that trigonometric functions have to the circle, as such they are collectively called Hyperbolic Functions and individually called **hyperbolic sine**, **hyperbolic cosine** and so on.

DEFINITION OF THE HYPERBOLIC FUNCTIONS

$\sinh x = \frac{e^x - e^{-x}}{2}$	$\operatorname{csch} x = \frac{1}{\sinh x}$
$\cosh x = \frac{e^x + e^{-x}}{2}$	$\operatorname{sech} x = \frac{1}{\cosh x}$
$\tanh x = \frac{\sinh x}{\cosh x}$	$\operatorname{coth} x = \frac{\cosh x}{\sinh x}$

HYPERBOLIC IDENTITIES

$$\begin{aligned}
 \sinh(-x) &= -\sinh x \\
 \cosh^2 x - \sinh^2 x &= 1 \\
 \cosh(-x) &= \cosh x \\
 1 &= \tanh^2 x + \operatorname{sech}^2 x
 \end{aligned}$$

Example 167 Show that $\cosh^2 t - \sinh^2 t = 1$

Solution 168 Exercise.

DERIVATIVES OF HYPERBOLIC FUNCTIONS

Let u be a differentiable function of x , then

$\frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$	$\frac{d}{dx}(\operatorname{csch} u) = -\operatorname{csch} u \coth u \frac{du}{dx}$
$\frac{d}{dx}(\cosh u) = \sinh u \frac{du}{dx}$	$\frac{d}{dx}(\operatorname{sech} u) = -\operatorname{sech} u \tanh u \frac{du}{dx}$
$\frac{d}{dx}(\tanh u) = (\operatorname{sech} u)^2 \frac{du}{dx}$	$\frac{d}{dx}(\coth u) = -(\operatorname{csch} u)^2 \frac{du}{dx}$

Example 169 Differentiate

$$y = \coth\left(\sqrt{x^2 + 1}\right)$$

Solution 170 Exercise.

Since the hyperbolic functions are defined in terms of the natural exponential functions, it seems natural that the inverse hyperbolic functions should be expressible in terms of natural logarithmic functions.

$\sinh^{-1} x = \ln\left(x + \sqrt{x^2 + 1}\right), x \in \mathbb{R}$
$\cosh^{-1} x = \ln\left(x + \sqrt{x^2 - 1}\right), x \geq 1$
$\tanh^{-1} x = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), -1 < x < 1$

Example 171 Show that $\sinh^{-1} x = \ln\left(x + \sqrt{x^2 + 1}\right)$.

Solution 172 Let $y = \sinh^{-1} x$. Then

$$x = \sinh y = \frac{e^y - e^{-y}}{2}$$

so

$$e^y - 2x - e^{-y} = 0$$

or

$$(e^y)^2 - 2x(e^y) - 1 = 0.$$

This is a quadratic equation in terms of e^y . From the quadratic formula, we get

$$e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2} = x \pm \sqrt{x^2 + 1}$$

We observe that $e^y > 0$, but $x - \sqrt{x^2 + 1} < 0$ (because $x < \sqrt{x^2 + 1}$). Thus

$$e^y = x + \sqrt{x^2 + 1}.$$

Therefore

$$y = \ln(e^y) = \ln(x + \sqrt{x^2 + 1})$$

Which shows that

$$\sinh^{-1} x = \ln \left(x + \sqrt{x^2 + 1} \right).$$

The other two are left as exercises.

DERIVATIVES OF INVERSE HYPERBOLIC FUNCTIONS

Let u be a differentiable function of x , then

$\frac{d}{dx} (\sinh^{-1} u) = \frac{1}{\sqrt{u^2 + 1}} \frac{du}{dx}$	$\frac{d}{dx} (\csc h^{-1} u) = \frac{-1}{ u \sqrt{u^2 + 1}} \frac{du}{dx}$
$\frac{d}{dx} (\cosh^{-1} u) = \frac{1}{\sqrt{u^2 - 1}} \frac{du}{dx}$	$\frac{d}{dx} (\sec h^{-1} u) = \frac{-1}{u \sqrt{1 - u^2}} \frac{du}{dx}$
$\frac{d}{dx} (\tanh^{-1} u) = \frac{1}{1 - u^2} \frac{du}{dx}$	$\frac{d}{dx} (\coth^{-1} u) = \frac{1}{1 - u^2} \frac{du}{dx}$

Example 173 Find the derivative of

$$y = x^2 \operatorname{sech}^{-1} 3x$$

Solution 174 Exercise.

Exercise 175 Find the derivative and simplify where possible.

1. $y = e^x \cosh x$
2. $y = \tanh \sqrt{x}$
3. $y = x \tanh^{-1} x + \ln \sqrt{1 - x^2}$
4. $y = \tanh^2 3x$
5. $y = (\cosh x)^{1/x}$
6. $f(x) = (x^2 + 1)^{\sinh x}$
7. Show that $\frac{d}{dx} \sqrt[4]{\frac{1 + \tanh x}{1 - \tanh x}} = \frac{1}{2} e^{x/2}$.

4 Applications

4.1 L'Hôpital's Rule

Theorem 176 Suppose that f and g are differentiable on an open interval I that contains c with the possible exception of c itself, and $g'(x) \neq 0$ for all $x \in I$. If

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$$

is an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then

$$\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$$

provided that the limit on the right hand side exists or is infinite.

Example 177 Evaluate

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{2x}$$

Solution 178 We have an indeterminate form of the type $\frac{0}{0}$. Applying L'Hôpital's Rule, we obtain

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \lim_{x \rightarrow 0} \frac{\frac{d}{dx}(e^x - 1)}{\frac{d}{dx}(2x)} = \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$$

Example 179 Evaluate

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x}$$

Solution 180 Exercise.

4.1.1 The Indeterminate form $\infty - \infty$ and $0 \cdot \infty$

If $\lim_{x \rightarrow c} f(x) = \infty$ and $\lim_{x \rightarrow c} g(x) = \infty$ then the limit

$$\lim_{x \rightarrow c} [f(x) - g(x)]$$

is said to be an indeterminate form of the type $\infty - \infty$. An indeterminate form of this type can often be expressed as one of the type $\frac{0}{0}$ or $\frac{\infty}{\infty}$ by algebraic manipulation.

Example 181 Evaluate

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right)$$

Solution 182 We have indeterminate form of the type $\infty - \infty$. By writing the expression in the parenthesis as a single fraction, we obtain an indeterminate form of the type $\frac{0}{0}$.

$$\begin{aligned} \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right) &= \lim_{x \rightarrow 0^+} \frac{e^x - x - 1}{x(e^x - 1)} \quad \left(\text{Type } \frac{0}{0} \right) \\ &= \lim_{x \rightarrow 0^+} \frac{e^x - 1}{(e^x - 1 + xe^x)} \quad (\text{Apply L'Hôpital's Rule}) \\ &= \lim_{x \rightarrow 0^+} \frac{e^x}{(x + 2)e^x} = \frac{1}{2} \end{aligned}$$

Example 183 Evaluate

$$\lim_{x \rightarrow 0^+} (x \ln x)$$

Solution 184 We have an indeterminate form of the type $0 \cdot \infty$. By writing

$$x \ln x = \frac{\ln x}{\frac{1}{x}}$$

the given limit can be cast in an indeterminate form of the type $\frac{\infty}{\infty}$. Then by applying L'Hôpital's Rule, we obtain

$$\begin{aligned} \lim_{x \rightarrow 0^+} (x \ln x) &= \lim_{x \rightarrow 0^+} \left(\frac{\ln x}{\frac{1}{x}} \right) \quad \left(\text{Type } \frac{\infty}{\infty} \right) \\ &= \lim_{x \rightarrow 0^+} \left(\frac{\frac{1}{x}}{-\frac{1}{x^2}} \right) \\ &= \lim_{x \rightarrow 0^+} (-x) = 0. \end{aligned}$$

4.1.2 The Indeterminate Forms 0^0 , ∞^0 and 1^∞

The limit

$$\lim_{x \rightarrow c} [f(x)]^{g(x)}$$

is said to be an indeterminate form of the type

$$\begin{aligned} 0^0 &\text{ if } \lim_{x \rightarrow c} f(x) = 0 \text{ and } \lim_{x \rightarrow c} g(x) = 0 \\ \infty^0 &\text{ if } \lim_{x \rightarrow c} f(x) = \infty \text{ and } \lim_{x \rightarrow c} g(x) = 0 \\ 1^\infty &\text{ if } \lim_{x \rightarrow c} f(x) = 1 \text{ and } \lim_{x \rightarrow c} g(x) = \pm\infty \end{aligned}$$

Example 185 Evaluate

$$\lim_{x \rightarrow 0^+} x^x$$

Solution 186 We have an indeterminate form of the type 0^0 . Let

$$y = x^x$$

Then

$$\ln y = x \ln x$$

From the last example we obtain

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln x = 0$$

Finally, using the identity

$$y = e^{\ln y}$$

and the continuity of the exponential function, we have

$$\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} e^{\ln y} = e^{\lim_{x \rightarrow 0^+} \ln y} = e^0 = 1$$

Example 187 Evaluate

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} \right)^{\sin x}$$

Solution 188 We have an indeterminate form of the type ∞^0 . Let

$$y = \left(\frac{1}{x} \right)^{\sin x}$$

Then

$$\ln y = \ln \left(\frac{1}{x} \right)^{\sin x} = \sin x \ln \left(\frac{1}{x} \right)$$

and

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} (\sin x) \ln \left(\frac{1}{x} \right)$$

which is an indeterminate form of the type $0 \cdot \infty$. By writing

$$\sin x \ln \left(\frac{1}{x} \right) = \frac{\ln \left(\frac{1}{x} \right)}{\frac{1}{\sin x}}$$

we can transform it into an indeterminate form of the type $\frac{\infty}{\infty}$ and hence use L'Hôpital's

Rule. We have

$$\begin{aligned}
 \lim_{x \rightarrow 0^+} \ln y &= \lim_{x \rightarrow 0^+} \left(\frac{\ln \left(\frac{1}{x} \right)}{\frac{1}{\sin x}} \right) \\
 &= \lim_{x \rightarrow 0^+} \left(\frac{-\ln x}{\frac{1}{\sin x}} \right) \\
 &= \lim_{x \rightarrow 0^+} \left(\frac{-\frac{1}{x}}{-\frac{\cos x}{\sin^2 x}} \right) \\
 &= \lim_{x \rightarrow 0^+} \frac{\sin^2 x}{x \cos x} \\
 &= \lim_{x \rightarrow 0^+} \left(\frac{\sin x}{x} \right) (\tan x) = (1) \cdot (0) = 0
 \end{aligned}$$

Therefore,

$$\lim_{x \rightarrow 0^+} \left(\frac{1}{x} \right)^{\sin x} = \lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} e^{\ln y} = e^{\lim_{x \rightarrow 0^+} \ln y} = e^0 = 1$$

Since the exponential function is continuous.

Example 189 Evaluate

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x$$

Solution 190 We have the indeterminate form of the type 1^∞ . Let

$$y = \left(1 + \frac{1}{x} \right)^x$$

Then

$$\ln y = \ln \left(1 + \frac{1}{x} \right)^x = x \ln \left(1 + \frac{1}{x} \right)$$

so

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} \left[x \ln \left(1 + \frac{1}{x} \right) \right]$$

has an indeterminate form of the type $0 \cdot \infty$. Rewriting and using L'Hôpital's Rule, we

obtain

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \ln y &= \lim_{x \rightarrow \infty} \left[x \ln \left(1 + \frac{1}{x} \right) \right] \\
 &= \lim_{x \rightarrow \infty} \left[\frac{\ln \left(1 + \frac{1}{x} \right)}{\frac{1}{x}} \right] \\
 &= \lim_{x \rightarrow \infty} \left[\frac{\left(\frac{1}{1 + \frac{1}{x}} \right) \left(-\frac{1}{x^2} \right)}{-\frac{1}{x^2}} \right] \\
 &= \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{1}{x}} = 1
 \end{aligned}$$

Therefore,

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = \lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} e^{\ln y} = e^{\lim_{x \rightarrow \infty} \ln y} = e^1 = e$$

since the exponential function is continuous.

Exercise 191 Find the Limits

1. $\lim_{x \rightarrow 0} \frac{\sin^{-1} 5x}{x}$
2. $\lim_{x \rightarrow 1} \frac{\ln x}{x-1}$
3. $\lim_{x \rightarrow 0} \frac{x2^x}{2^x - 1}$
4. $\lim_{x \rightarrow \infty} \frac{\ln(x+1)}{\log_2 x}$
5. $\lim_{x \rightarrow \infty} (\ln 2x - \ln(x+1))$
6. $\lim_{x \rightarrow \infty} \frac{(1+e^x)}{x^2}$
7. $\lim_{x \rightarrow \infty} x^2 e^{-x}$
8. $\lim_{x \rightarrow \infty} (1+2x)^{1/(2 \ln x)}$
9. $\lim_{x \rightarrow 0^+} x^x$
10. $\lim_{x \rightarrow \infty} \left(\frac{x+2}{x-1} \right)^x$
11. $\lim_{x \rightarrow 0^+} x^2 \ln x$
12. $\lim_{x \rightarrow 0^+} x (\ln x)^2$
13. $\lim_{x \rightarrow 0^+} \left(1 + \frac{1}{x} \right)^x$
14. $\lim_{x \rightarrow 1^+} x^{1/(x-1)}$
15. $\lim_{x \rightarrow \infty} x^{1/\ln x}$

4.2 Taylor and Maclaurin Polynomials

Definition 192 (nth Taylor and Maclaurin Polynomials) If f has derivatives at a , then the polynomial

$$\begin{aligned}
 P_n(x) &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 \\
 &\quad + \frac{f^{(4)}(a)}{4!}(x-a)^4 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n
 \end{aligned}$$

is called the n th **Taylor polynomial** for f at a . If $a = 0$, then

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(n)}(0)}{n!}x^n$$

is called the n th **Maclaurin polynomial** for f .

Example 193 Find the Maclaurin polynomial for $f(x) = \sin x$

$$f(x) = \sin x \quad f(0) = 0$$

$$f'(x) = \cos x \quad f'(0) = 1$$

$$f''(x) = -\sin x \quad f''(0) = 0$$

$$f'''(x) = -\cos x \quad f'''(0) = -1$$

$$f^{(4)}(x) = \sin x \quad f^{(4)}(0) = 0$$

$$\begin{aligned} P_n(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots + \frac{f^{(n)}(0)}{n!}x^n \\ &= x - \frac{1}{3!}x^3 + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots + \frac{f^{(n)}(0)}{n!}x^n \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}x^{2n+1} \end{aligned}$$

Example 194 Find the Taylor polynomials P_0, P_1, P_2, P_3 and P_4 for $f(x) = \ln x$ centered at $a = 1$.

$$f(x) = \ln x \quad f(1) = 0$$

$$f'(x) = \frac{1}{x} \quad f'(1) = 1$$

$$f''(x) = -\frac{1}{x^2} \quad f''(1) = -1$$

$$f'''(x) = \frac{2!}{x^3} \quad f'''(1) = 2$$

$$f^{(4)}(x) = -\frac{3!}{x^4} \quad f^{(4)}(1) = -6$$

Therefore, the Taylor polynomials are as follows,

$$P_0(x) = f(1) = 0$$

$$P_1(x) = f(1) + f'(1)(x-1) = (x-1)$$

$$P_2(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 = (x-1) - \frac{1}{2}(x-1)^2$$

$$\begin{aligned} P_3(x) &= f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3 \\ &= (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 \end{aligned}$$

$$\begin{aligned}
 P_4(x) &= f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3 + \frac{f^{(4)}(1)}{4!}(x-1)^4 \\
 &= (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4
 \end{aligned}$$

Exercise 195 Find the Maclaurin polynomial of degree n for the function

1. $f(x) = e^{-x}$, $n = 3$
2. $f(x) = e^{3x}$, $n = 4$
3. $f(x) = \ln(1+x)$, $n = 4$

Find the Maclaurin series for

4. $f(x) = \cos x$
5. $f(x) = e^x$
6. $f(x) = \sinh x$

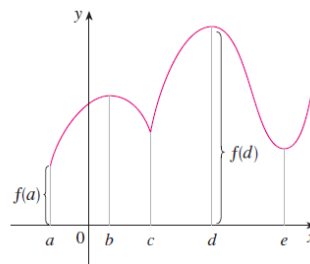
Find the n th Taylor polynomial centered at a

7. $f(x) = \frac{1}{x}$, $n = 4$, $a = 1$
8. $f(x) = \ln x$, $n = 4$, $a = 1$

4.3 Maxima and Minima

Definition 196 Let c be a number in the domain D of a function f . Then $f(c)$ is the

- **absolute maximum** value of f on D if $f(c) \geq f(x)$ for all $x \in D$.
- **absolute minimum** value of f on D if $f(c) \leq f(x)$ for all $x \in D$.



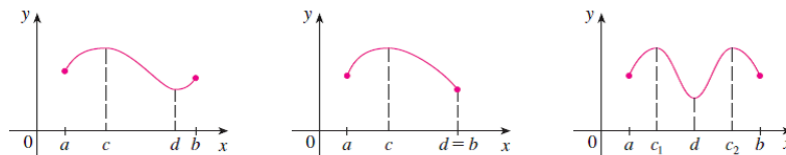
Extrema: Abs min $f(a)$, abs max $f(d)$, loc min $f(c), f(e)$, loc max $f(b), f(d)$.

An absolute maximum or minimum is sometimes called a **global** maximum or minimum. The maximum or minimum values of f are called **extreme values** of f .

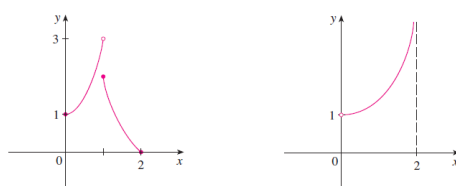
Definition 197 The number $f(c)$ is a

- **local maximum** value of f if $f(c) \geq f(x)$ when x is near c .
- **local minimum** value of f if $f(c) \leq f(x)$ when x is near c .

Theorem 198 (Extreme Value Theorem) *If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.*



A function need not possess extreme values if either hypotheses (continuity or closed interval) is omitted from the Extreme Value Theorem.



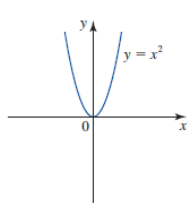
Example 199 Find the extrema of the function if any by examining the given graphs.

(a) $f(x) = x^2$.

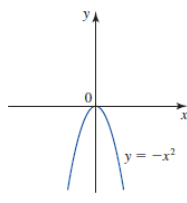
(b) $g(x) = -x^2$.

(c) $h(x) = \frac{1}{x}$.

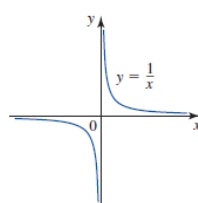
(d) $k(x) = \frac{x}{\sqrt{x^2 + 7}}$.



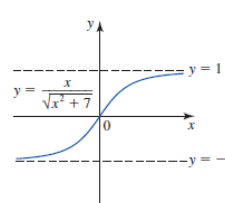
(a) min at 0



(b) max at 0



(c) no extrema



(d) no extrema

(a) f has a minimum value of 0 at 0. Since the values of f are not bounded above, f has no maximum value.

(b) g has a maximum value of 0 at 0. Also, because the values of g are not bounded below, g has no minimum value.

(c) The values of h are neither bounded above nor bounded below, so h has no absolute extrema.

- (d) As x gets larger and larger, $k(x)$ gets closer and closer to 1 but the value is never attained; that is a real number c does not exist such that $k(c) = 1$. Therefore k has no maximum value. Similarly k has no minimum value.

The Extreme Value Theorem says that a continuous function on a closed interval has a maximum value and a minimum value, but it does not tell us how to find these extreme values. We note that at the maximum and minimum points the tangent lines are horizontal and thus each has a gradient (slope) equal to 0.

Theorem 200 (Fermat's Theorem) *If f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$.*

Fermat's Theorem suggests that we should at least start to look for extreme values of f at the number c where $f'(c) = 0$ or $f'(c)$ does not exist.

Definition 201 A **critical number** of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

Example 202 Find the critical numbers of $f(x) = x^{3/5}(4-x)$.

Solution 203 The product rule gives

$$\begin{aligned} f'(x) &= x^{3/5}(-1) + (4-x)\left(\frac{3}{5}x^{-2/5}\right) \\ &= -x^{3/5} + \frac{3(4-x)}{5x^{2/5}} = \frac{-5x + 12 - 3x}{5x^{2/5}} = \frac{12 - 8x}{5x^{2/5}} \end{aligned}$$

Therefore $f'(x) = 0$ if $12 - 8x = 0$, that is $x = \frac{3}{2}$, and $f'(x)$ does not exist when $x = 0$.

Thus the critical numbers are $\frac{3}{2}$ and 0.

Example 204 Find the critical number(s) if any, of $f(x) = x - 3x^{1/3}$.

Solution 205 Exercise

If f has a local maximum or minimum at c , then c is a critical number of f .

THE CLOSED INTERVAL METHOD To find the *absolute* maximum and minimum (extreme) values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical numbers of f in (a, b) .
2. Find the values of f at the endpoints of the interval.
3. The largest of the values from steps 1 and 2 is the absolute maximum value; the smallest of these is the absolute minimum value.

Example 206 Find the extreme values of the function $f(x) = 2\cos x - x$ on $[0, 2\pi]$

Solution 207 The function f is continuous everywhere, in particular it is continuous on the closed interval $[0, 2\pi]$. Therefore the Extreme Value Theorem is applicable. First we find the critical numbers of f in $(0, 2\pi)$. We have

$$f'(x) = -\sin x - 1$$

Observe that f' is continuous on $(0, 2\pi)$. Setting $f'(x) = 0$ gives

$$\begin{aligned} -2\sin x - 1 &= 0 \\ \sin x &= -\frac{1}{2} \end{aligned}$$

Reference angle:

$$\alpha = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}.$$

But $\sin \alpha$ is negative in the 3rd and 4th quadrants. Thus $x = \pi + \alpha = \pi + \frac{\pi}{6} = \frac{7\pi}{6}$ or $x = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$.

Next we compute the values of f at these critical numbers as well as at the end points 0 and 2π .

x	0	$\frac{7\pi}{6}$	$\frac{11\pi}{6}$	2π
$f(x)$	2	-5.40	-4.03	-4.28

From the table we see that f attains the absolute maximum value of 2 at 0 and the absolute minimum value of approximately -5.4 at $7\pi/6$.

Exercise 208 Find the extreme (absolute maximum and absolute minimum) values, if any, of the function.

1. $f(x) = x^2 - x - 2$ on $[0, 2]$
2. $f(x) = 2x^4 - \frac{8}{3}x^3 - 8x^2 + 12$ on $[-2, 3]$
3. $f(x) = \frac{x}{x^2 + 1}$ on $[-1, 2]$
4. $f(x) = 2 + 3\sin 2x$ on $[0, \frac{\pi}{2}]$

5. The average speed of traffic flow from Sibasa to Thohoyandou between 6 A.M. and 10 A.M. on a typical weekday is approximated by the function

$$f(t) = 20t - 40\sqrt{t} + 50 \quad 0 \leq t \leq 4$$

where $f(t)$ is measured in kilometers per hour and t is measured in hours, with $t = 0$ corresponding to 6 A.M. At what time in the morning is the average speed of traffic flow highest? At what time in the morning is it lowest?

6. A body of mass m moves in an elliptical path with a constant angular speed ω . The force acting on the body is always directed toward the origin and has the magnitude given by

$$F = m\omega^2 \sqrt{a^2 \cos^2 \omega t + b^2 \sin^2 \omega t} \quad t \geq 0$$

where a and b are constants with $a > b$. Find the points on the path where the force is greatest and where it is smallest.

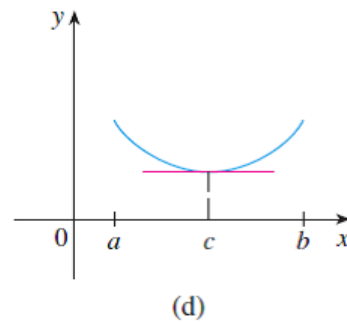
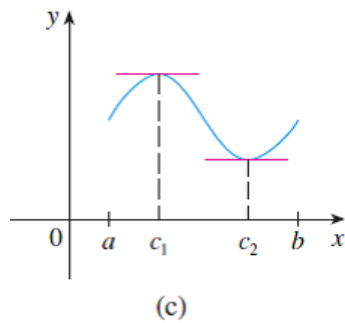
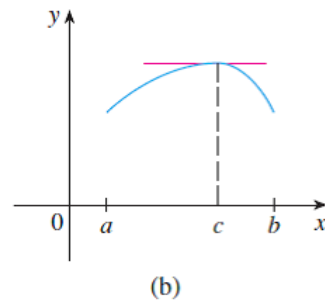
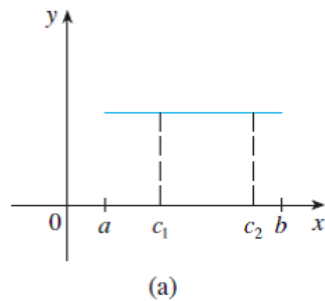
4.4 The Mean Value Theorem

We shall prove the main result of this chapter. To do so we need the following result.

Theorem 209 (Rolle's Theorem) *Let f be a function that satisfies the following three hypotheses:*

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .
3. $f(a) = f(b)$.

Then there is a number c in (a, b) such that $f'(c) = 0$.



Example 210 *Let $f(x) = x^3 - x$ for $x \in [-1, 1]$.*

1. *Show that f satisfies the hypotheses of Rolle's Theorem on $[-1, 1]$.*
2. *Find the numbers c in $(-1, 1)$ such that $f'(c) = 0$ as guaranteed by Rolle's Theorem.*

Theorem 211 (The Mean Value Theorem) *Let f be a function that satisfies the following hypotheses:*

1. f is continuous on the closed interval $[a, b]$.

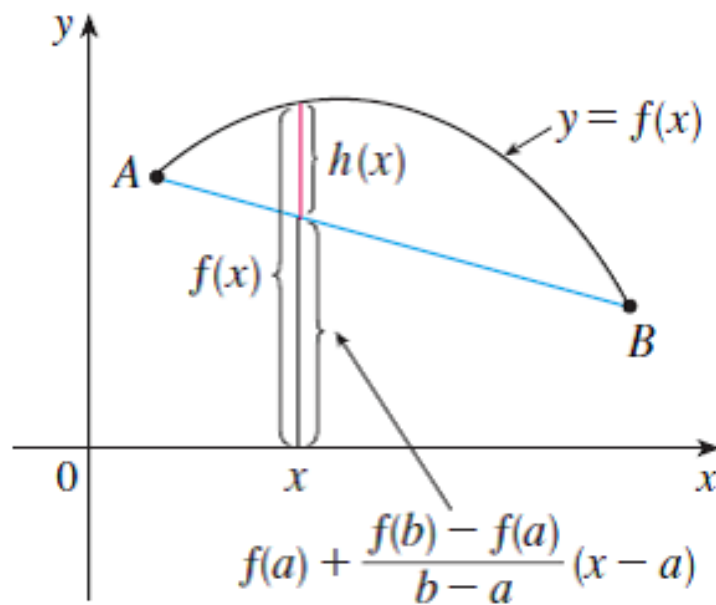
2. f is differentiable on the open interval (a, b) .

Then there is a number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

or equivalently,

$$f(b) - f(a) = f'(c)(b - a).$$



Proof. From the figure above, the gradient of the secant line AB is given by

$$\frac{f(b) - f(a)}{b - a}$$

The equation of the secant line AB is :

$$\begin{aligned} y - f(a) &= \frac{f(b) - f(a)}{b - a}(x - a) \\ y &= \frac{f(b) - f(a)}{b - a}(x - a) + f(a) \end{aligned}$$

Let $h(x)$ be the difference between $f(x)$ and y .

$$h(x) = f(x) - y$$

This implies that

$$h(x) = f(x) - \left[\frac{f(b) - f(a)}{b - a}(x - a) + f(a) \right]$$

But

$$h(a) = h(b) = 0$$

Furthermore, because f is differentiable, h is also differentiable on (a, b) , and we apply Rolle's Theorem to the function h . Thus there exists a number c in (a, b) such that $h'(c) = 0$. Since

$$h'(x) = f'(x) - \left[\frac{f(b) - f(a)}{b - a} \right]$$

we have

$$h'(c) = f'(c) - \left[\frac{f(b) - f(a)}{b - a} \right]$$

Since

$$\begin{aligned} h'(c) &= 0, \\ f'(c) &= \frac{f(b) - f(a)}{b - a} \end{aligned}$$

■

Example 212 Find the number(s) c that satisfies the conclusion of the MVT on the given interval

$$f(x) = x^3 - x, [0, 2]$$

Solution 213 Since f is a polynomial it is continuous and differentiable for all x , so it is certainly continuous on $[0, 2]$ and differentiable on $(0, 2)$. By the MVT there is a number c in $(0, 2)$ such that

$$f(2) - f(0) = f'(c)(2 - 0)$$

Now $f(2) = 6$, $f(0) = 0$, and $f'(x) = 3x^2 - 1$, so the equation becomes

$$6 = (3c^2 - 1)2 = 6c^2 - 2$$

which gives $c^2 = \frac{4}{3}$, that is $c = \pm \frac{2}{\sqrt{3}}$. But c must lie in $(0, 2)$, so $c = \frac{2}{\sqrt{3}}$.

4.5 First Derivative Test

Definition 214 A function f is **increasing** on an interval I , if for every pair of numbers x_1 and x_2 in I ,

$$x_1 < x_2 \text{ implies that } f(x_1) < f(x_2)$$

f is **decreasing** on I , if for every pair of numbers x_1 and x_2 in I ,

$$x_1 < x_2 \text{ implies that } f(x_1) > f(x_2)$$

f is **monotonic** on I , if it is either increasing or decreasing on I ,

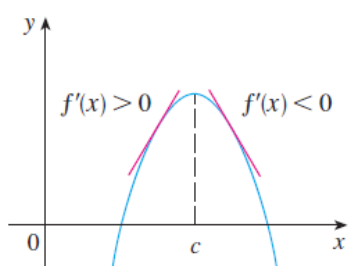
Increasing /Decreasing Test
(a) If $f'(x) > 0$ on an interval, then f is increasing on that interval.
(b) If $f'(x) < 0$ on an interval, then f is decreasing on that interval.

The First Derivative Test Suppose that c is a critical number of a continuous function f .

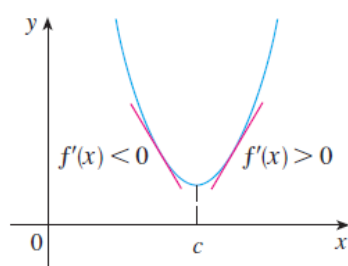
(a) If f' changes from the positive to negative at c , then f has a local maximum at c .

(b) If f' changes from the negative to positive at c , then f has a local minimum at c .

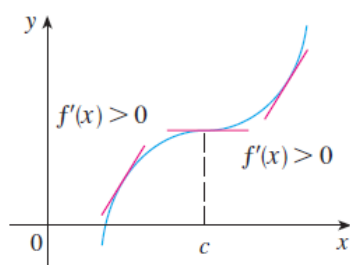
(c) If f' does not change sign at c (e.g. f' is positive on both sides of c or negative on both sides), then f has no local maximum or minimum at c .



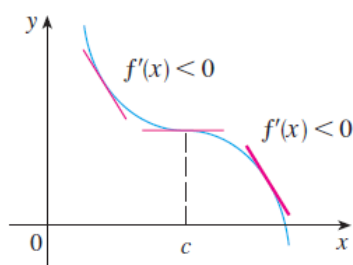
(a) Local maximum



(b) Local minimum

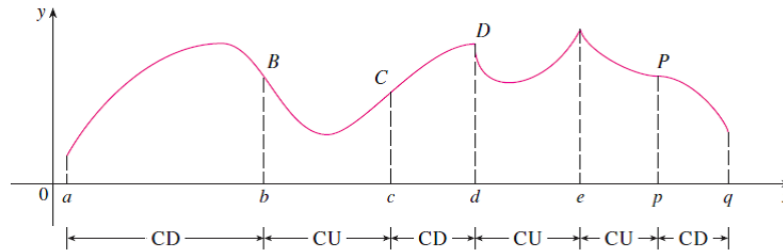


(c) No maximum or minimum



(d) No maximum or minimum

Definition 215 If the graph of f lies above all of its tangents on an interval I , then it is called **concave upward** on I . If the graph of f lies below all of its tangents on I , it is called **concave downward** on I .



Concavity Test

- (a) If $f''(x) > 0$ for all x on I , then the graph of f is concave upward on I .
- (b) If $f''(x) < 0$ for all x on I , then the graph of f is concave downward on I .

Definition 216 A point P on the curve $y = f(x)$ is called an **inflection point** if f is continuous there and the curve changes from concave upward to concave downward or from concave downward to concave upward at P .

Definition 217 (The Second Derivative Test) Suppose f'' is continuous near c ,

- (a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c ,
- (b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .

4.6 Curve Sketching

Guidelines for Curve Sketching

1. Find the domain of f .
2. Find the x - and y - intercepts of f .
3. Determine whether the graph of f is symmetric with respect to the y - axis or the origin.
4. Determine the behaviour of f for large absolute values of x .
5. Find the asymptotes of the graph of f .
6. Find the intervals where f is increasing and where f is decreasing.
7. Find the relative extrema of f .
8. Determine the concavity of the graph of f .
9. Find the inflection points of f .
10. Sketch the graph of f .

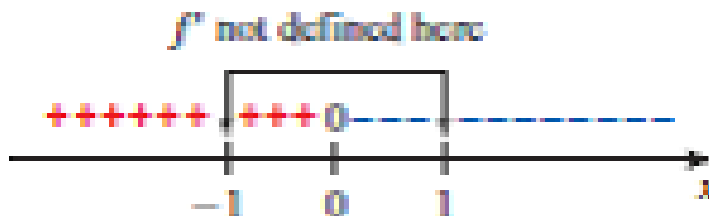
Example 218 Sketch the graph of the function $f(x) = \frac{x^2}{x^2 - 1}$.

1. The denominator of the rational function is equal to zero if $x^2 - 1 = (x + 1)(x - 1) = 0$, that is if $x = -1$ or $x = 1$. Therefore, the domain of f is $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$.
2. Setting $x = 0$ gives 0 as y -intercept. Next setting $f(x) = 0$ gives $x = 0$ as the x -intercept.
3. $f(-x) = \frac{(-x)^2}{(-x)^2 - 1} = \frac{x^2}{x^2 - 1} = f(x)$ and this shows that f is an even function and thus the graph of f is symmetric with respect to the y -axis.
4. $\lim_{x \rightarrow -\infty} \frac{x^2}{x^2 - 1} = \lim_{x \rightarrow \infty} \frac{x^2}{x^2 - 1} = 1$. Thus $y = 1$ is the horizontal asymptote.
5. Because the denominator of $f(x)$ is equal to 0 at -1 and 1 , the lines $x = -1$ and $x = 1$ are candidates of the vertical asymptotes of the graph of f . Since

$$\lim_{x \rightarrow -1^-} \frac{x^2}{x^2 - 1} = \infty \text{ and } \lim_{x \rightarrow 1^-} \frac{x^2}{x^2 - 1} = -\infty$$

we see that $x = -1$ and $x = 1$ are the vertical asymptotes.

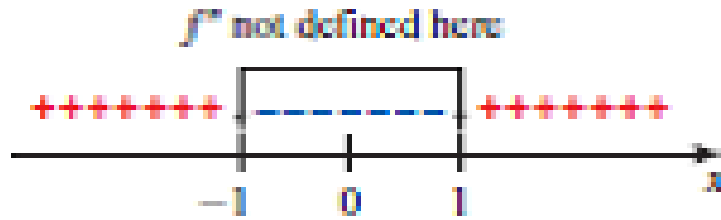
6. $f'(x) = \frac{(x^2 - 1)(2x) - x^2(2x)}{(x^2 - 1)^2} = -\frac{2x}{(x^2 - 1)^2}$. Notice that f' is continuous everywhere except at ± 1 and that it has a zero when $x = 0$. The sign diagram of f' is shown below.



From the diagram we see that f is increasing on $(-\infty, -1)$ and $(-1, 0)$ and decreasing on $(0, 1)$ and on $(1, \infty)$.

7. From the results of part (6) we see that 0 is a critical number of f . The numbers -1 and 1 are not in the domain of f , and therefore, are not critical numbers of f . Also, from the sign diagram we see that f has a relative maximum at $x = 0$. Its value is $f(0) = 0$.

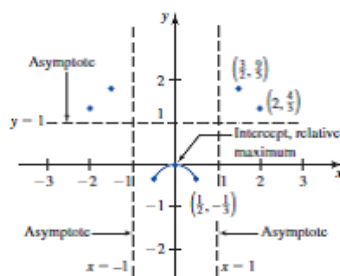
8. $f''(x) = \frac{d}{dx} \left[\frac{-2x}{(x^2 - 1)^2} \right] = \frac{2(3x^2 + 1)}{(x^2 - 1)^3}$. Notice that f'' is continuous everywhere except at ± 1 and that f'' has no zeros. From the sign diagram of f'' we see that the graph of f is concave upward on $(-\infty, -1)$ and on $(1, \infty)$ and concave downward on $(-1, 1)$.



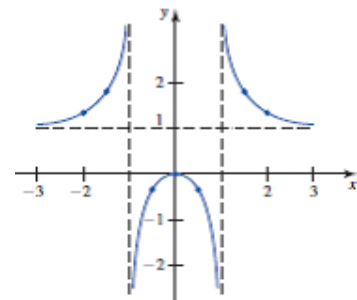
9. f has no inflection points. Remember that -1 and 1 are not in the domain of f .

10. The following table summarizes this information

Domain	$(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$
Intercepts	x - and y -intercepts: 0
Symmetry	With respect to y -axis
Asymptotes	Vertical: $x = -1$ and $x = 1$
	Horizontal: $y = 1$
End Behaviour	$\lim_{x \rightarrow -\infty} \frac{x^2}{x^2 - 1} = \lim_{x \rightarrow \infty} \frac{x^2}{x^2 - 1} = 1$
Intervals where f is $\nearrow$ or $\searrow$	$\nearrow$ on $(-\infty, -1)$ and $(-1, 0)$; $\searrow$ on $(0, 1)$ and $(1, \infty)$
Relative extrema	Rel max at $(0, 0)$
Concavity	$\downarrow$ on $(-1, 1)$; $\uparrow$ on $(-\infty, -1)$ and on $(1, \infty)$
Point of inflection	None



(a)



(b)

Exercise 219 Use the guidelines given above to sketch the graphs of the following functions

1. $f(x) = 2x^3 - 3x^2 - 12x + 12$

2. $f(x) = \frac{x}{x+1}$

3. $f(x) = -\frac{8}{x^2 - 4}$

4. $y = \frac{x^2 - 4}{x^2 - 2x}$

5. $y = \sqrt{x^2 + x - 2}$

6. $y = \sin^3 x$

